

Chapter IV

Abstract Class Field Theory

§ 1. Infinite Galois Theory

Every field k is equipped with a distinguished Galois extension: the separable closure $\bar{k}|k$. Its Galois group $G_k = G(\bar{k}|k)$ is called the **absolute Galois group** of k . As a rule, this extension will have infinite degree. It does, however, have the advantage of collecting all finite Galois extensions of k . This is why it is reasonable to try to give it a prominent place in Galois theory. But such an attempt faces the difficulty that the main theorem of Galois theory does not remain true for infinite extensions. Let us explain this in the following

Example: The absolute Galois group $G_{\mathbb{F}_p} = G(\bar{\mathbb{F}}_p|\mathbb{F}_p)$ of the field $\mathbb{F}_p$ with p elements contains the Frobenius automorphism φ which is given by

$$x^\varphi = x^p \quad \text{for all } x \in \bar{\mathbb{F}}_p.$$

The subgroup $(\varphi) = \{\varphi^n \mid n \in \mathbb{Z}\}$ has the same fixed field $\mathbb{F}_p$ as the whole of $G_{\mathbb{F}_p}$. But contrary to what we are used to in finite Galois theory, we find $(\varphi) \neq G_{\mathbb{F}_p}$. In order to check this, let us construct an element $\psi \in G_{\mathbb{F}_p}$ which does not belong to (φ) . We choose a sequence $\{a_n\}_{n \in \mathbb{N}}$ of integers satisfying

$$a_n \equiv a_m \pmod{m}$$

whenever $m|n$, but such that there is no integer a satisfying $a_n \equiv a \pmod{n}$ for all $n \in \mathbb{N}$. An example of such a sequence is given by $a_n = n'x_n$, where we write $n = n'p^{v_p(n)}$, $(n', p) = 1$, and $1 = n'x_n + p^{v_p(n)}y_n$. Now put

$$\psi_n = \varphi^{a_n}|_{\mathbb{F}_{p^n}} \in G(\mathbb{F}_{p^n}|\mathbb{F}_p).$$

If $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$, then $m|n$, so that $a_n \equiv a_m \pmod{m}$, and therefore

$$\psi_n|_{\mathbb{F}_{p^m}} = \varphi^{a_n}|_{\mathbb{F}_{p^m}} = \varphi^{a_m}|_{\mathbb{F}_{p^m}} = \psi_m.$$

Observe that $\varphi|_{\mathbb{F}_{p^m}}$ has order m . Therefore the ψ_n define an automorphism ψ of $\bar{\mathbb{F}}_p = \bigcup_{n=1}^{\infty} \mathbb{F}_{p^n}$. Now ψ cannot belong to (φ) because $\psi = \varphi^a$, for $a \in \mathbb{Z}$, would imply $\psi|_{\mathbb{F}_{p^n}} = \varphi^{a_n}|_{\mathbb{F}_{p^n}} = \varphi^a|_{\mathbb{F}_{p^n}}$ and hence $a_n \equiv a \pmod{n}$ for all n , which is what we ruled out by construction.

The example does not mean, however, that we have to chuck the main theorem of Galois theory altogether in the case of infinite extensions. We just have to amend it using the observation that the Galois group $G = G(\Omega|k)$ of any Galois extension $\Omega|k$ carries a canonical topology. This topology is called the **Krull topology** and is obtained as follows. For every $\sigma \in G$ we take the cosets

$$\sigma G(\Omega|K)$$

as a basis of neighbourhoods of σ , with $K|k$ ranging over *finite* Galois subextensions of $\Omega|k$. The multiplication and the inverse map

$$G \times G \longrightarrow G, (\sigma, \tau) \longmapsto \sigma\tau, \quad \text{and} \quad G \longrightarrow G, \sigma \longmapsto \sigma^{-1},$$

are continuous maps, since the preimage of a fundamental open neighbourhood $\sigma\tau G(\Omega|K)$, resp. $\sigma^{-1}G(\Omega|K)$, contains the open neighbourhood $\sigma G(\Omega|K) \times \tau G(\Omega|K)$, resp. $\sigma G(\Omega|K)$. Thus G is a topological group which satisfies the following

(1.1) Proposition. *For every (finite or infinite) Galois extension $\Omega|k$ the Galois group $G = G(\Omega|k)$ is compact Hausdorff with respect to the Krull topology.*

Proof: If $\sigma, \tau \in G$ and $\sigma \neq \tau$, then there exists a finite Galois subextension $K|k$ of $\Omega|k$ such that $\sigma|_K \neq \tau|_K$, so that $\sigma G(\Omega|K) \neq \tau G(\Omega|K)$ and thus $\sigma G(\Omega|K) \cap \tau G(\Omega|K) = \emptyset$. This shows that G is Hausdorff. In order to prove compactness, consider the mapping

$$h : G \longrightarrow \prod_K G(K|k), \quad \sigma \longmapsto \prod_K \sigma|_K,$$

where $K|k$ varies over the finite Galois subextensions. We view the finite groups $G(K|k)$ as discrete compact topological groups. Their product is therefore a compact topological space, by Tykhonov's theorem (see [98]). The homomorphism h is injective, because $\sigma|_K = 1$ for all K is equivalent to $\sigma = 1$. The sets $U = \prod_{K \neq K_0} G(K|k) \times \{\bar{\sigma}\}$ form a subbasis of open sets of the product $\prod_K G(K|k)$, where $K_0|k$ varies over the finite subextensions of $\Omega|k$ and $\bar{\sigma} \in G(K_0|k)$. If $\sigma \in G$ is a preimage of $\bar{\sigma}$, then $h^{-1}(U) = \sigma G(\Omega|K_0)$. Thus h is continuous. Moreover $h(\sigma G(\Omega|K_0)) = h(G) \cap U$, so $h : G \rightarrow h(G)$ is open, and thus a homeomorphism. It therefore suffices to show that $h(G)$ is closed in the compact set $\prod_K G(K|k)$. To see this we consider, for each pair $L' \supseteq L$ of finite Galois subextensions of $\Omega|k$, the set

$$M_{L'|L} = \left\{ \prod_K \sigma_K \in \prod_K G(K|k) \mid \sigma_{L'}|_L = \sigma_L \right\}.$$

One clearly has $h(G) = \bigcap_{L' \supseteq L} M_{L'|L}$. So it suffices to show that $M_{L'|L}$ is closed. But if $G(L|k) = \{\sigma_1, \dots, \sigma_n\}$, and $S_i \subseteq G(L'|k)$ is the set of extensions of σ_i to L' , then

$$M_{L'|L} = \bigcup_{i=1}^n \left(\prod_{K \neq L, L'} G(K|k) \times S_i \times \{\sigma_i\} \right),$$

i.e., $M_{L'|L}$ is indeed closed. □

The main theorem of Galois theory for infinite extensions can now be formulated as follows.

(1.2) Theorem. *Let $\Omega|k$ be a (finite or infinite) Galois extension. Then the assignment*

$$K \longmapsto G(\Omega|K)$$

is a 1–1-correspondence between the subextensions $K|k$ of $\Omega|k$ and the closed subgroups of $G(\Omega|k)$. The open subgroups of $G(\Omega|k)$ correspond precisely to the finite subextensions of $\Omega|k$.

Proof: Every open subgroup of $G(\Omega|k)$ is also closed, because it is the complement of the union of its open cosets. If $K|k$ is a finite subextension, then $G(\Omega|K)$ is open, because each $\sigma \in G(\Omega|K)$ admits the open neighbourhood $\sigma G(\Omega|N) \subseteq G(\Omega|K)$, where $N|k$ is the normal closure of $K|k$. If $K|k$ is an arbitrary subextension, then

$$G(\Omega|K) = \bigcap_i G(\Omega|K_i),$$

where $K_i|k$ varies over the finite subextensions of $K|k$. Therefore $G(\Omega|K)$ is closed.

The assignment $K \mapsto G(\Omega|K)$ is injective, since K is the fixed field of $G(\Omega|K)$. To prove surjectivity, we have to show that, given an arbitrary closed subgroup H of $G(\Omega|k)$, we always have

$$H = G(\Omega|K),$$

where K is the fixed field of H . The inclusion $H \subseteq G(\Omega|K)$ is trivial. Conversely, let $\sigma \in G(\Omega|K)$. If $L|K$ is a finite Galois subextension of $\Omega|K$, then $\sigma G(\Omega|L)$ is a fundamental open neighbourhood of σ in $G(\Omega|K)$. The map $H \rightarrow G(L|K)$ is certainly surjective, because the image $\overline{H}$ has fixed field K and is therefore equal to $G(L|K)$, by the main theorem of Galois theory for finite extensions. Thus we may choose a $\tau \in H$ such that

$\tau|_L = \sigma|_L$, i.e., $\tau \in H \cap \sigma G(\Omega|L)$. This shows that σ belongs to the closure of H in $G(\Omega|K)$, and thus to H itself, so that $H = G(\Omega|K)$.

If H is an open subgroup of $G(\Omega|k)$, then it is also closed, and therefore of the form $H = G(\Omega|K)$. But $G(\Omega|k)$ is the disjoint union of the open cosets of H . Since $G(\Omega|k)$ is compact, a finite number of cosets suffices to cover the group. Thus there is only a finite number of them; $H = G(\Omega|K)$ has finite index in $G(\Omega|k)$, and this implies that $K|k$ has finite degree. $\square$

The topological Galois groups $G = G(\Omega|k)$ have the special property that there is a fundamental system of neighbourhoods of the neutral element $1 \in G$ which consists of normal subgroups. This property leads us to the abstract, purely group-theoretical notion of a profinite group.

(1.3) Definition. A **profinite group** is a topological group G which is Hausdorff and compact, and which admits a basis of neighbourhoods of $1 \in G$ consisting of normal subgroups.

It can be shown that the last condition is tantamount to G being totally disconnected, i.e., to the condition that each element of G is equal to its own connected component. Every closed subgroup H of G is obviously again a profinite group. The disjoint coset decomposition

$$G = \bigcup_i \sigma_i H$$

shows immediately that H is open if and only if the index $(G : H)$ is finite.

Profinite groups are fairly close relatives of finite groups. They can be reconstituted rather easily from their finite quotients. For the precise description of this we need the notion of *projective limit*, which naturally occurs in various places in number theory and which we will introduce next.

Exercise 1. Let $L|k$ be a Galois extension and $K|k$ an arbitrary extension, both contained in a common extension $\Omega|k$. If $L \cap K = k$, then the mapping

$$G(LK|K) \rightarrow G(L|k), \quad \sigma \mapsto \sigma|_L,$$

is a topological isomorphism, that is, an isomorphism of groups and a homeomorphism of topological spaces.

Exercise 2. Given a family of Galois extensions $K_i|k$ in $\Omega|k$, let $K|k$ be the composite of all $K_i|k$, and $K'_j|k$ the composite of the extensions $K_j|k$ such that $j \neq i$. If $K_i \cap K'_i = k$ for all i , then one has a topological isomorphism

$$G(K|k) \cong \prod_i G(K_i|k).$$

Exercise 3. A compact Hausdorff group is totally disconnected if and only if its neutral element admits a basis of neighbourhoods consisting only of normal subgroups.

Exercise 4. Every quotient G/H of a profinite group G by a closed normal subgroup H is a profinite group.

Exercise 5. Let G' be the closure of the commutator subgroup of a profinite group, and $G^{ab} = G/G'$. Show that every continuous homomorphism $G \rightarrow A$ into an abelian profinite group factorizes through G^{ab} .

§ 2. Projective and Inductive Limits

The notions of projective, resp. inductive limit generalize the operations of intersection, resp. union. If $\{X_i\}_{i \in I}$ is a family of subsets of a topological space X which for any two sets X_i, X_j also contains the set $X_i \cap X_j$ (resp. $X_i \cup X_j$), then the projective (resp. inductive) limit of this family is simply defined by

$$\varprojlim_{i \in I} X_i = \bigcap_{i \in I} X_i \quad (\text{resp.} \quad \varinjlim_{i \in I} X_i = \bigcup_{i \in I} X_i).$$

Writing $i \leq j$ if $X_j \subseteq X_i$ (resp. $X_i \subseteq X_j$) makes the indexing set I into a *directed system*, i.e., an ordered set in which, for every pair i, j , there exists a k such that $i \leq k$ and $j \leq k$. In the case at hand, such a k is given by $X_k = X_i \cap X_j$ (resp. $X_k = X_i \cup X_j$). For $i \leq j$ we denote the inclusion $X_j \hookrightarrow X_i$ (resp. $X_i \hookrightarrow X_j$) by f_{ij} and obtain a system $\{X_i, f_{ij}\}$ of sets and maps. The operations of intersection and union are now generalized by replacing the inclusions f_{ij} with arbitrary maps.

(2.1) Definition. Let I be a directed system. A **projective, resp. inductive system** over I is a family $\{X_i, f_{ij} \mid i, j \in I, i \leq j\}$ of topological spaces X_i and continuous maps

$$f_{ij} : X_j \longrightarrow X_i, \quad \text{resp.} \quad f_{ij} : X_i \longrightarrow X_j,$$

such that one has $f_{ii} = \text{id}_{X_i}$ and

$$f_{ik} = f_{ij} \circ f_{jk}, \quad \text{resp.} \quad f_{ik} = f_{jk} \circ f_{ij},$$

when $i \leq j \leq k$.

In order to define the projective, resp. inductive limit of a projective, resp. inductive system $\{X_i, f_{ij}\}$, we make use of the direct product $\prod_{i \in I} X_i$, resp. the disjoint union $\coprod_{i \in I} X_i$.

(2.2) Definition. *The projective limit*

$$X = \varprojlim_{i \in I} X_i$$

of the projective system $\{X_i, f_{ij}\}$ is defined to be the subset

$$X = \left\{ (x_i)_{i \in I} \in \prod_{i \in I} X_i \mid f_{ij}(x_j) = x_i \text{ for } i \leq j \right\}$$

of the product $\prod_{i \in I} X_i$.

The product $\prod_{i \in I} X_i$ is equipped with the product topology. If the X_i are Hausdorff, then so is the product, and it contains in this case X as a closed subspace. Indeed, one has

$$X = \bigcap_{i \leq j} X_{ij},$$

where $X_{ij} = \{(x_k)_{k \in I} \in \prod_k X_k \mid f_{ij}(x_j) = x_i\}$, so that it suffices to show the closedness of the sets X_{ij} . Writing $p_i : \prod_{k \in I} X_k \rightarrow X_i$ for the i -th projection, the two maps $g = p_i$, $f = f_{ij} \circ p_j : \prod_{k \in I} X_k \rightarrow X_i$ are continuous, and we may write $X_{ij} = \{x \in \prod_k X_k \mid g(x) = f(x)\}$. But in the Hausdorff case the equation $g(x) = f(x)$ defines a closed subset. This representation $X = \bigcap_{i \leq j} X_{ij}$ also gives the following

(2.3) Proposition. *The projective limit $X = \varprojlim_i X_i$ of nonempty compact spaces X_i is itself nonempty and compact.*

Proof: If all the X_i are compact, then so is the product $\prod_{i \in I} X_i$, by Tychonov's theorem, and thus also the closed subset X . Furthermore, $X = \bigcap_{i \leq j} X_{ij}$ cannot be the empty set if the X_i are nonempty. In fact, as the product $\prod_i X_i$ is compact, there would have to be an intersection of finitely many X_{ij} which is empty. But this is impossible: if all indices entering into this finite intersection satisfy $i, j \leq n$, and if $x_n \in X_n$, then the element $(x_i)_{i \in I}$ belongs to this intersection, where we choose $x_i = f_{in}(x_n)$, for $i \leq n$, and arbitrarily for all other i . $\square$

(2.4) Definition. *The inductive limit*

$$X = \varinjlim_{i \in I} X_i$$

of an inductive system $\{X_i, f_{ij}\}$ is defined to be the quotient

$$X = (\coprod_{i \in I} X_i) / \sim$$

of the disjoint union $\coprod_{i \in I} X_i$, where we consider two elements $x_i \in X_i$ and $x_j \in X_j$ equivalent if there exists a $k \geq i, j$ such that

$$f_{ik}(x_i) = f_{jk}(x_j).$$

In the applications, the projective and inductive systems $\{X_i, f_{ij}\}$ that occur will not just be systems of topological spaces and continuous maps, but the X_i will usually be topological groups, rings or modules, etc., and the f_{ij} will be continuous homomorphisms. In what follows, we will deal explicitly only with projective and inductive systems $\{G_i, g_{ij}\}$ of topological groups. But since everything works exactly the same way for systems of rings or modules, these cases may be thought of tacitly as being treated as well.

Let $\{G_i, g_{ij}\}$ be a projective, resp. inductive system of topological groups. Then the projective, resp. inductive limit

$$G = \varprojlim_{i \in I} G_i, \quad \text{resp.} \quad G = \varinjlim_{i \in I} G_i$$

is a topological group as well. The multiplication in the projective limit is induced by the componentwise multiplication in the product $\prod_{i \in I} G_i$. In the case of the inductive limit, given two equivalence classes $x, y \in G = \varinjlim_{i \in I} G_i$, one has to choose representatives x_k and y_k in the same G_k in order to define

$$xy = \text{equivalence class of } x_k y_k.$$

We leave it to the reader to check that this definition is independent of the choice of representatives, and that the operation thus defined makes G into a group.

The projections $p_i : \prod_{i \in I} G_i \rightarrow G_i$, resp. the inclusions $\iota_i : G_i \rightarrow \prod_{i \in I} G_i$, induce a family of continuous homomorphisms

$$g_i : G \rightarrow G_i, \quad \text{resp.} \quad g_i : G_i \rightarrow G$$

such that $g_i = g_{ij} \circ g_j$, resp. $g_i = g_j \circ g_{ij}$, for $i \leq j$. This family has the following universal property.

(2.5) Proposition. *If H is a topological group and*

$$h_i : H \rightarrow G_i, \quad \text{resp.} \quad h_i : G_i \rightarrow H$$

is a family of continuous homomorphisms such that

$$h_i = g_{ij} \circ h_j, \quad \text{resp.} \quad h_i = h_j \circ g_{ij}$$

for $i \leq j$, then there exists a unique continuous homomorphism

$$h : H \rightarrow G = \varprojlim_i G_i, \quad \text{resp.} \quad h : G = \varinjlim_i G_i \rightarrow H$$

satisfying $h_i = g_i \circ h$, resp. $h_i = h \circ g_i$, for all $i \in I$.

The easy proof is left to the reader. A **morphism** between two projective, resp. inductive systems $\{G_i, g_{ij}\}$ and $\{G'_i, g'_{ij}\}$ of topological groups is a family of continuous homomorphisms $f_i : G_i \rightarrow G'_i$, $i \in I$, such that the diagrams

$$\begin{array}{ccc} G_j & \xrightarrow{f_j} & G'_j \\ g_{ij} \downarrow & & \downarrow g'_{ij} \\ G_i & \xrightarrow{f_i} & G'_i \end{array} \quad \text{resp.} \quad \begin{array}{ccc} G_j & \xrightarrow{f_j} & G'_j \\ g_{ij} \uparrow & & \uparrow g'_{ij} \\ G_i & \xrightarrow{f_i} & G'_i \end{array}$$

commute for $i \leq j$. Such a family $(f_i)_{i \in I}$ defines a mapping

$$f : \prod_{i \in I} G_i \longrightarrow \prod_{i \in I} G'_i, \quad \text{resp.} \quad f : \coprod_{i \in I} G_i \longrightarrow \coprod_{i \in I} G'_i,$$

which induces a homomorphism

$$f : \varprojlim_{i \in I} G_i \longrightarrow \varprojlim_{i \in I} G'_i, \quad \text{resp.} \quad f : \varinjlim_{i \in I} G_i \longrightarrow \varinjlim_{i \in I} G'_i.$$

In this way $\varprojlim$, resp. $\varinjlim$, becomes a functor. A particularly important property of this functor is its so-called “exactness”. For the inductive limit $\varinjlim$, exactness holds without restrictions. In other words, one has the

(2.6) Proposition. *Let $\alpha : \{G'_i, g'_{ij}\} \rightarrow \{G_i, g_{ij}\}$ and $\beta : \{G_i, g_{ij}\} \rightarrow \{G''_i, g''_{ij}\}$ be morphisms between inductive systems of topological groups such that the sequence*

$$G'_i \xrightarrow{\alpha_i} G_i \xrightarrow{\beta_i} G''_i$$

is exact for every $i \in I$. Then the induced sequence

$$\varinjlim_{i \in I} G'_i \xrightarrow{\alpha} \varinjlim_{i \in I} G_i \xrightarrow{\beta} \varinjlim_{i \in I} G''_i$$

is also exact.

Proof: Let $G' = \varinjlim_i G'_i$, $G = \varinjlim_i G_i$, $G'' = \varinjlim_i G''_i$. We consider the commutative diagram

$$\begin{array}{ccccc} G'_i & \xrightarrow{\alpha_i} & G_i & \xrightarrow{\beta_i} & G''_i \\ \downarrow g'_i & & \downarrow g_i & & \downarrow g''_i \\ G' & \xrightarrow{\alpha} & G & \xrightarrow{\beta} & G'' \end{array}$$

Let $x \in G$ be such that $\beta(x) = 1$. Then there exists an i and an $x_i \in G_i$ such that $g_i(x_i) = x$. As

$$g_i'' \beta_i(x_i) = \beta g_i(x_i) = \beta(x) = 1,$$

there exists $j \geq i$ such that $\beta_j(x_i)$ equals 1 in G_j'' . Changing notation, we may therefore assume that $\beta_i(x_i) = 1$, so that there exists $y_i \in G_i'$ such that $\alpha_i(y_i) = x_i$. Putting $y = g_i'(y_i)$, we have $\alpha(y) = x$. $\square$

The projective limit is not exact in complete generality, but only for **compact** groups, so that we have the

(2.7) Proposition. *Let $\alpha : \{G_i', g_{ij}'\} \rightarrow \{G_i, g_{ij}\}$ and $\beta : \{G_i, g_{ij}\} \rightarrow \{G_i'', g_{ij}''\}$ be morphisms between projective systems of **compact** topological groups such that the sequence*

$$G_i' \xrightarrow{\alpha_i} G_i \xrightarrow{\beta_i} G_i''$$

is exact for every $i \in I$. Then

$$\varprojlim_i G_i' \xrightarrow{\alpha} \varprojlim_i G_i \xrightarrow{\beta} \varprojlim_i G_i''$$

is again an exact sequence of compact topological groups.

Proof: Let $x = (x_i)_{i \in I} \in \varprojlim_i G_i$ and $\beta(x) = 1$, so that $\beta_i(x_i) = 1$ for all $i \in I$. The preimages $Y_i = \alpha_i^{-1}(x_i) \subseteq G_i'$ then form a projective system of nonempty closed, and hence compact subsets of the G_i' . By (2.3), this means that the projective limit $Y = \varprojlim_i Y_i \subseteq \varprojlim_i G_i'$ is nonempty, and α maps every element $y \in Y$ to x . $\square$

Now that we have at our disposal the notion of projective limit, we return to our starting point, the profinite groups. Recall that these are the topological groups which are Hausdorff, compact and totally disconnected, i.e., they admit a basis of neighbourhoods of the neutral element consisting of normal subgroups. The next proposition shows that they are precisely the projective limits of finite groups (which we view as compact topological groups with respect to the discrete topology).

(2.8) Proposition. *If G is a profinite group, and if N varies over the open normal subgroups of G , then one has, algebraically as well as topologically, that*

$$G \cong \varprojlim_N G/N.$$

If conversely $\{G_i, g_{ij}\}$ is a projective system of finite (or even profinite) groups, then

$$G = \varprojlim_i G_i$$

is a profinite group.

Proof: Let G be a profinite group and let $\{N_i \mid i \in I\}$ be the family of its open normal subgroups. We make I into a directed system by defining $i \leq j$ if $N_i \supseteq N_j$. The groups $G_i = G/N_i$ are finite since the cosets of N_i in G form a disjoint open covering of G , which must be finite because G is compact. For $i \leq j$ we have the projections $g_{ij} : G_j \rightarrow G_i$ and obtain a projective system $\{G_i, g_{ij}\}$ of finite, and hence discrete, compact groups. We show that the homomorphism

$$f : G \longrightarrow \varprojlim_{i \in I} G_i, \quad \sigma \longmapsto \prod_{i \in I} \sigma_i, \quad \sigma_i = \sigma \bmod N_i,$$

is an isomorphism and a homeomorphism. f is injective because its kernel is the intersection $\bigcap_{i \in I} N_i$, which equals $\{1\}$ because G is Hausdorff and the N_i form a basis of neighbourhoods of 1. The groups

$$U_S = \prod_{i \notin S} G_i \times \prod_{i \in S} \{1_{G_i}\},$$

with S varying over the finite subsets of I , form a basis of neighbourhoods of the neutral element in $\prod_{i \in I} G_i$. As $f^{-1}(U_S \cap \varprojlim G_i) = \bigcap_{i \in S} N_i$, we see that f is continuous. Moreover, as G is compact, the image $f(G)$ is closed in $\varprojlim G_i$. On the other hand it is also dense. For if $x = (x_i)_{i \in I} \in \varprojlim G_i$, and $x(U_S \cap \varprojlim G_i)$ is a fundamental neighbourhood of x , then we may choose a $y \in G$ which is mapped to x_k under the projection $G \rightarrow G/N_k$, where we put $N_k = \bigcap_{i \in S} N_i$. Then $y \bmod N_i = x_i$ for all $i \in S$, so that $f(y)$ belongs to the neighbourhood $x(U_S \cap \varprojlim G_i)$. Therefore the closed set $f(G)$ is indeed dense in $\varprojlim G_i$, and so $f(G) = \varprojlim G_i$. Since G is compact, f maps closed sets into closed sets, and thus also open sets into open sets. This shows that $f : G \rightarrow \varprojlim G_i$ is an isomorphism and a homeomorphism.

Conversely, let $\{G_i, g_{ij}\}$ be a projective system of profinite groups. As the G_i are Hausdorff and compact, so is the projective limit $G = \varprojlim G_i$,

by (2.3). If N_i varies over a basis of neighbourhoods of the neutral element in G_i which consists of normal subgroups, then the groups

$$U_S = \prod_{i \notin S} G_i \times \prod_{i \in S} N_i,$$

with S varying over the finite subsets of I , make up a basis of neighbourhoods of the neutral element in $\prod_{i \in I} G_i$ consisting of normal subgroups. The normal subgroups $U_S \cap \varprojlim G_i$ therefore form a basis of neighbourhoods of the neutral element in $\varprojlim G_i$; thus $\varprojlim G_i$ is a profinite group. $\square$

Let us now illustrate the notions of profinite group and projective limit by a few concrete examples.

Example 1: The Galois group $G = G(\Omega|k)$ of a Galois extension $\Omega|k$ is a profinite group with respect to the Krull topology. This was already stated in §1. If $K|k$ varies over the finite Galois subextensions of $\Omega|k$, then, by definition of the Krull topology, $G(\Omega|K)$ varies over the open normal subgroups of G . In view of the identity $G(K|k) = G(\Omega|k)/G(\Omega|K)$ and of (2.8), we therefore obtain the Galois group $G(\Omega|k)$ as the projective limit

$$G(\Omega|k) \cong \varprojlim G(K|k)$$

of the finite Galois groups $G(K|k)$.

Example 2: If p is a prime number, then the rings $\mathbb{Z}/p^n\mathbb{Z}$, $n \in \mathbb{N}$, form a projective system with respect to the projections $\mathbb{Z}/p^n\mathbb{Z} \rightarrow \mathbb{Z}/p^m\mathbb{Z}$, for $n \geq m$. The projective limit

$$\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$$

is the ring of p -adic integers (see chap. II, §1).

Example 3: Let $\mathfrak{o}$ be the valuation ring in a p -adic number field K and $\mathfrak{p}$ its maximal ideal. The ideals $\mathfrak{p}^n$, $n \in \mathbb{N}$, make up a basis of neighbourhoods of the zero element 0 in $\mathfrak{o}$. $\mathfrak{o}$ is Hausdorff and compact, and so is a profinite ring. The rings $\mathfrak{o}/\mathfrak{p}^n$, $n \in \mathbb{N}$, are finite and we have a topological isomorphism

$$\mathfrak{o} \cong \varprojlim_n \mathfrak{o}/\mathfrak{p}^n, \quad a \mapsto \prod_{n \in \mathbb{N}} (a \bmod \mathfrak{p}^n).$$

The group of units $U = \mathfrak{o}^*$ is closed in $\mathfrak{o}$, hence Hausdorff and compact, and the subgroups $U^{(n)} = 1 + \mathfrak{p}^n$ form a basis of neighbourhoods of $1 \in U$. Thus

$$U \cong \varprojlim_n U/U^{(n)}$$

is also a profinite group. In fact, we have seen all this already in chap. II, §4.

Example 4: The rings $\mathbb{Z}/n\mathbb{Z}$, $n \in \mathbb{N}$, form a projective system with respect to the projections $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$, $n|m$, where the ordering on $\mathbb{N}$ is now given by divisibility, $n|m$. The projective limit

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$$

was originally called the **Prüfer ring**, whereas nowadays it has become customary to refer to it by the somewhat curt abbreviation “zed-hat” (or “zee-hat”). This ring is going to occupy quite an important position in what follows. It contains $\mathbb{Z}$ as a dense subring. The groups $n\widehat{\mathbb{Z}}$, $n \in \mathbb{N}$, are precisely the open subgroups of $\widehat{\mathbb{Z}}$, and it is easy to verify that

$$\widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}.$$

Taking, for each natural number n , the prime factorization $n = \prod_p p^{v_p}$, the Chinese remainder theorem implies the decomposition

$$\mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}/p^{v_p}\mathbb{Z},$$

and passing to the projective limit,

$$\widehat{\mathbb{Z}} \cong \prod_p \mathbb{Z}_p.$$

This takes the natural embedding of $\mathbb{Z}$ into $\widehat{\mathbb{Z}}$ to the diagonal embedding $\mathbb{Z} \rightarrow \prod_p \mathbb{Z}_p$, $a \mapsto (a, a, a, \dots)$.

Example 5: For the field $\mathbb{F}_q$ with q elements, we get isomorphisms

$$G(\mathbb{F}_{q^n}|\mathbb{F}_q) \cong \mathbb{Z}/n\mathbb{Z},$$

one for every $n \in \mathbb{N}$, by mapping the Frobenius automorphism φ_n to $1 \bmod n\mathbb{Z}$. Passing to the projective limit gives an isomorphism

$$G(\overline{\mathbb{F}}_q|\mathbb{F}_q) \cong \widehat{\mathbb{Z}}$$

which sends the Frobenius automorphism $\varphi \in G(\overline{\mathbb{F}}_q|\mathbb{F}_q)$ to $1 \in \widehat{\mathbb{Z}}$, and the subgroup $(\varphi) = \{\varphi^n \mid n \in \mathbb{Z}\}$ onto the dense (but not closed) subgroup $\mathbb{Z}$ of $\widehat{\mathbb{Z}}$. Given this, it is now clear, in the example at the beginning of this chapter, how we were able to construct an element $\psi \in G(\overline{\mathbb{F}}_q|\mathbb{F}_q)$ which did not belong to (φ) . In fact, looking at it via the isomorphism $G(\overline{\mathbb{F}}_q|\mathbb{F}_q) \cong \widehat{\mathbb{Z}}$, what we did amounted to writing down the element

$$(\dots, 0, 0, 1_p, 0, 0, \dots) \in \prod_\ell \mathbb{Z}_\ell = \widehat{\mathbb{Z}},$$

which does not belong to $\mathbb{Z}$.

Example 6: Let $\tilde{\mathbb{Q}}|\mathbb{Q}$ be the extension obtained by adjoining all roots of unity. Its Galois group $G(\tilde{\mathbb{Q}}|\mathbb{Q})$ is then canonically isomorphic (as a topological group) to the group of units $\widehat{\mathbb{Z}}^* \cong \prod_p \mathbb{Z}_p^*$ of $\widehat{\mathbb{Z}}$,

$$G(\tilde{\mathbb{Q}}|\mathbb{Q}) \cong \widehat{\mathbb{Z}}^*.$$

This isomorphism is obtained by passing to the projective limit from the canonical isomorphisms

$$G(\mathbb{Q}(\mu_n)|\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^*,$$

where μ_n denotes the group of n -th roots of unity.

Example 7: The groups $\mathbb{Z}_p$ and $\widehat{\mathbb{Z}}$ are (additive) special cases of the class of **procyclic** groups. These are profinite groups G which are topologically generated by a single element σ ; i.e., G is the closure $(\overline{\sigma})$ of the subgroup $(\sigma) = \{\sigma^n \mid n \in \mathbb{Z}\}$. The open subgroups of a procyclic group $G = (\overline{\sigma})$ are all of the form G^n . Indeed, G^n is closed, being the image of the continuous map $G \rightarrow G$, $x \mapsto x^n$, and the quotient group G/G^n is finite, because it contains the finite group $\{\sigma^\nu \bmod G^n \mid 0 \leq \nu < n\}$ as a dense subgroup, and is therefore equal to it. Conversely, if H is a subgroup of G of index n , then $G^n \subseteq H \subseteq G$ and $n = (G : H) \leq (G : G^n) \leq n$, so that $H = G^n$.

Every procyclic group G is a quotient of the group $\widehat{\mathbb{Z}}$. In fact, if $G = (\overline{\sigma})$, then we have for every n the surjective homomorphism

$$\mathbb{Z}/n\mathbb{Z} \rightarrow G/G^n, \quad 1 \bmod n\mathbb{Z} \mapsto \sigma \bmod G^n,$$

and in view of (2.7), passing to the projective limit yields a continuous surjection $\widehat{\mathbb{Z}} \rightarrow G$.

Example 8: Let A be an abelian torsion group. Then the **Pontryagin dual**

$$\chi(A) = \text{Hom}(A, \mathbb{Q}/\mathbb{Z})$$

is a profinite group. For one has

$$A = \bigcup_i A_i,$$

where A_i varies over the finite subgroups of A , and thus

$$\chi(A) = \varprojlim_i \chi(A_i)$$

with finite groups $\chi(A_i)$. If for instance,

$$A = \mathbb{Q}/\mathbb{Z} = \bigcup_{n \in \mathbb{N}} \frac{1}{n}\mathbb{Z}/\mathbb{Z},$$

then $\chi(\frac{1}{n}\mathbb{Z}/\mathbb{Z}) = \mathbb{Z}/n\mathbb{Z}$, so that

$$\chi(\mathbb{Q}/\mathbb{Z}) \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}.$$

Example 9: If G is any group and N varies over all normal subgroups of finite index, then the profinite group

$$\widehat{G} = \varprojlim_N G/N$$

is called the *profinite completion* of G . The profinite completion of $\mathbb{Z}$, for example, is the group $\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$.

Exercise 1. Show that, for a profinite group G , the power map $G \times \mathbb{Z} \rightarrow G$, $(\sigma, n) \mapsto \sigma^n$, extends to a continuous map

$$G \times \widehat{\mathbb{Z}} \rightarrow G, \quad (\sigma, a) \mapsto \sigma^a,$$

and that one has $(\sigma^a)^b = \sigma^{ab}$ and $\sigma^{a+b} = \sigma^a \sigma^b$ if G is abelian.

Exercise 2. If $\sigma \in G$ and $a = \lim_{i \rightarrow \infty} a_i \in \widehat{\mathbb{Z}}$ with $a_i \in \mathbb{Z}$, then $\sigma^a = \lim_{i \rightarrow \infty} \sigma^{a_i}$ is in G .

Exercise 3. A *pro- p -group* is a profinite group G whose quotients G/N , modulo all open normal subgroups N , are finite p -groups. Imitating exercise 1, make sense of the powers σ^a , for all $\sigma \in G$ and $a \in \mathbb{Z}_p$.

Exercise 4. A closed subgroup H of a profinite group G is called a **p -Sylow subgroup** of G if, for every open normal subgroup N of G , the group HN/N is a p -Sylow subgroup of G/N . Show:

- (i) For every prime number p , there exists a p -Sylow subgroup of G .
- (ii) Every pro- p -subgroup of G is contained in a p -Sylow subgroup.
- (iii) Every two p -Sylow subgroups of G are conjugate.

Exercise 5. What is the p -Sylow subgroup of $\widehat{\mathbb{Z}}$ and of $\mathbb{Z}_p^*$?

Exercise 6. If $\{G_i\}$ is a projective system of profinite groups and $G = \varprojlim_i G_i$, then $G^{ab} = \varprojlim_i G_i^{ab}$ (see § 1, exercise 5).

§ 3. Abstract Galois Theory

Class field theory is the final outcome of a long development of algebraic number theory the beginning of which was Gauss's reciprocity law

$$\left(\frac{a}{b}\right)\left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2} \frac{b-1}{2}}.$$

The endeavours to generalize this law finally produced a theory of the abelian extensions of algebraic and p -adic number fields. These extensions $L|K$ are classified by certain subgroups $\mathcal{N}_L = N_{L|K} A_L$ of a group A_K attached to the base field. In the local case, A_K is the multiplicative group K^* and in the global case it is a modification of the ideal class group. At the heart of this theory there is a mysterious canonical isomorphism

$$G(L|K) \cong A_K / N_{L|K} A_L,$$

which – if we view things in the right way – encapsulates the reciprocity law in its most general form. Now, this map can be abstracted completely from the field-theoretic situation and treated on a purely group theoretical basis. In this way, class field theory can be given an abstract, but elementary foundation, to which we will now turn.

We begin our considerations by giving ourselves a profinite group G . The theory we are about to develop is purely group theoretical in nature. However, the only applications we have in mind are field theoretical, and the language of field theory allows immediate insights into the group theoretical relations. We will therefore formally interpret the profinite group G as a Galois group in the following way. (Let us remark in passing that every profinite group is indeed the Galois group $G = G(\bar{k}|k)$ of a Galois field extension $\bar{k}|k$; this will allow the reader to rely on his standard knowledge of Galois theory whenever the formal development in terms of group theory alone would seem odd.)

We denote the closed subgroups of G by G_K , and call these indices K “fields”; K will be called the fixed field of G_K . The field k such that $G_k = G$ is called the base field, and $\bar{k}$ denotes the field satisfying $G_{\bar{k}} = \{1\}$. The field belonging to the closure $\overline{(\sigma)}$ of the cyclic group $(\sigma) = \{\sigma^k \mid k \in \mathbb{Z}\}$ generated by an element $\sigma \in G$ is simply called the fixed field of σ .

We write formally $K \subseteq L$ or $L|K$ if $G_L \subseteq G_K$, and we call the pair $L|K$ a field extension. $L|K$ is called a finite extension, if G_L is open, i.e., of finite index in G_K , and this index

$$[L : K] := (G_K : G_L)$$

will be called the degree of $L|K$. $L|K$ is said to be normal or Galois if G_L is a normal subgroup of G_K . If this is the case, we define the Galois group of $L|K$ by

$$G(L|K) = G_K / G_L.$$

If $N \supseteq L \supseteq K$ are Galois extensions of K , we define the restriction of an element $\sigma \in G(N|K)$ to L by

$$\sigma|_L = \sigma \bmod G(N|L) \in G(L|K).$$

This gives a homomorphism

$$G(N|K) \longrightarrow G(L|K), \quad \sigma \longmapsto \sigma|_L,$$

with kernel $G(N|L)$. The extension $L|K$ is called cyclic, abelian, solvable, etc., if the Galois group $G(L|K)$ has these properties. We put

$$K = \bigcap_i K_i \quad (\text{“intersection”})$$

if G_K is topologically generated by the subgroups G_{K_i} , and

$$K = \prod_i K_i \quad (\text{“composite”})$$

if $G_K = \bigcap_i G_{K_i}$. If $G_{K'} = \sigma^{-1} G_K \sigma$ for $\sigma \in G$, we write $K' = K^\sigma$.

Now let A be a *continuous multiplicative* G -module. By this we mean a multiplicative abelian group A on which the elements $\sigma \in G$ operate as automorphisms on the right, $\sigma : A \rightarrow A, a \mapsto a^\sigma$. This action must satisfy

- (i) $a^1 = a$,
- (ii) $(ab)^\sigma = a^\sigma b^\sigma$,
- (iii) $a^{\sigma^\tau} = (a^\sigma)^\tau$,
- (iv) $A = \bigcup_{[K:k] < \infty} A_K$,

where A_K in the last condition denotes the fixed module A^{G_K} under G_K , so that

$$A_K = \{a \in A \mid a^\sigma = a \text{ for all } \sigma \in G_K\},$$

and where K varies over all extensions that are finite over k . The condition (iv) says that G operates continuously on A , i.e., the map

$$G \times A \longrightarrow A, \quad (\sigma, a) \longmapsto a^\sigma,$$

is continuous, where A is equipped with the discrete topology. Indeed, this continuity is equivalent to the fact that, for every element $(\sigma, a) \in G \times A$, there exists an open subgroup $U = G_K$ of G such that the neighbourhood $\sigma U \times \{a\}$ of (σ, a) is mapped to the open set $\{a^\sigma\}$, and this means simply that $a^\sigma \in A^U = A_K$.

Remark: In the exponential notation a^σ , the operation of G on A appears as an action on the right. This notation is adequate for many computations in the case of multiplicative G -modules A . For instance, the notation $a^{\sigma^{-1}} := a^\sigma a^{-1}$ is to be preferred to writing $(\sigma - 1)a = \sigma a \cdot a^{-1}$. On the other hand, classical usage often calls for an operation on the left. Thus in the case of a Galois extension $L|K$ of actual fields, the Galois group $G(L|K)$ acts as the automorphism group on L from the left, and therefore also in the same way on the multiplicative group L^* . This occasional switch from the left to the right should not confuse the reader.

For every extension $L|K$ we have $A_K \subseteq A_L$, and if $L|K$ is finite, then we have the norm map

$$N_{L|K} : A_L \longrightarrow A_K, \quad N_{L|K}(a) = \prod_{\sigma} a^{\sigma},$$

where σ varies over a system of representatives of $G_L \setminus G_K$. If $L|K$ is Galois, then A_L is a $G(L|K)$ -module and one has

$$A_L^{G(L|K)} = A_K.$$

At the center of class field theory there is the **norm residue group**

$$H^0(G(L|K), A_L) = A_K / N_{L|K} A_L.$$

We also consider the group

$$H^{-1}(G(L|K), A_L) = N_{L|K} A_L / I_{G(L|K)} A_L,$$

where

$$N_{L|K} A_L = \{a \in A_L \mid N_{L|K}(a) = 1\}$$

is the “norm-one group” and $I_{G(L|K)} A_L$ is the subgroup of $N_{L|K} A_L$ which is generated by all elements

$$a^{\sigma^{-1}} := a^\sigma a^{-1},$$

with $a \in A_L$, and $\sigma \in G(L|K)$. If $G(L|K)$ is cyclic and σ is a generator, then $I_{G(L|K)} A_L$ is simply the group

$$A_L^{\sigma^{-1}} = \{a^{\sigma^{-1}} \mid a \in A_L\}.$$

In fact, the formal identity $\sigma^k - 1 = (1 + \sigma + \cdots + \sigma^{k-1})(\sigma - 1)$ implies $a^{\sigma^k - 1} = b^{\sigma - 1}$ with $b = \prod_{i=0}^{k-1} a^{\sigma^i}$.

Let us now apply the notions introduced so far to the example of **Kummer theory**. For this, we impose on the G -module A the following axiomatic condition.

(3.1) Axiom. One has $H^{-1}(G(L|K), A_L) = 1$ for all finite cyclic extensions $L|K$.

The theory we are about to develop makes reference to a surjective G -homomorphism

$$\wp : A \longrightarrow A, \quad a \longmapsto a^\wp,$$

with finite cyclic kernel $\mu_\wp$. The order $n = \#\mu_\wp$ is called the *exponent* of the operator $\wp$. The case of prime interest to us is when $\wp$ is the n -th power map $a \mapsto a^n$, and $\mu_\wp = \mu_n = \{\xi \in A \mid \xi^n = 1\}$ is the group of “ n -th roots of unity” in A .

We now fix a field K such that $\mu_\wp \subseteq A_K$. For every subset $B \subseteq A$, let $K(B)$ denote the fixed field of the closed subgroup

$$H = \{\sigma \in G_K \mid b^\sigma = b \text{ for all } b \in B\}$$

of G_K . If B is G_K -invariant, then $K(B)|K$ is obviously Galois. A **Kummer extension** (with respect to $\wp$) is by definition an extension of the form

$$K(\wp^{-1}(\Delta)) \mid K,$$

where $\Delta \subseteq A_K$. A Kummer extension $K(\wp^{-1}(\Delta))|K$ is always Galois, and its Galois group is abelian of exponent n . Indeed, for an extension $K(\wp^{-1}(a))|K$, we have the injective homomorphism

$$G(K(\wp^{-1}(a))|K) \hookrightarrow \mu_\wp, \quad \sigma \longmapsto \alpha^{\sigma-1},$$

where $\alpha \in \wp^{-1}(a)$. Since $\mu_\wp \subseteq A_K$, this definition does not depend on the choice of α . Thus, for a Kummer extension $L = K(\wp^{-1}(\Delta)) = \prod_{a \in \Delta} K(\wp^{-1}(a))$, the composite map

$$G(L|K) \longrightarrow \prod_{a \in \Delta} G(K(\wp^{-1}(a))|K) \longrightarrow \mu_\wp^\Delta$$

is an injective homomorphism.

The following proposition says that conversely, any abelian extension $L|K$ of exponent n is a Kummer extension.

(3.2) Proposition. If $L|K$ is an abelian extension of exponent n , then

$$L = K(\wp^{-1}(\Delta)) \quad \text{with} \quad \Delta = A_L^\wp \cap A_K.$$

If in particular, $L|K$ is cyclic, then we find $L = K(\alpha)$ with $\alpha^\wp = a \in A_K$.

Proof: We have $\wp^{-1}(\Delta) \subseteq A_L$, for if $x \in A$ and $x^\wp = \alpha^\wp = a \in A_K$, $\alpha \in A_L$, then $x = \xi\alpha \in A_L$ for some $\xi \in \mu_\wp \subseteq A_K$. Therefore $K(\wp^{-1}(\Delta)) \subseteq L$. On the other hand, the extension $L|K$ is the composite of its cyclic subextensions. For it is the composite of its *finite* subextensions, and the Galois group of a finite subextension is the product of cyclic groups, which may be interpreted as Galois groups of cyclic subextensions. Let now $M|K$ be a cyclic subextension of $L|K$. It suffices to show that $M \subseteq K(\wp^{-1}(\Delta))$. Let σ be a generator of $G(M|K)$ and ζ a generator of $\mu_\wp$. Let $d = [M : K]$, $d' = n/d$ and $\xi = \zeta^{d'}$. Since $N_{M|K}(\xi) = \xi^d = 1$, (3.1) shows that $\xi = \alpha^{\sigma^{-1}}$ for some $\alpha \in A_M$. Thus $K \subseteq K(\alpha) \subseteq M$. But $\alpha^{\sigma^i} = \xi^i \alpha$. Thus $\alpha^{\sigma^i} = \alpha$ is equivalent to $i \equiv 0 \pmod{d}$, so that $K(\alpha) = M$. But $(\alpha^\wp)^{\sigma^{-1}} = (\alpha^{\sigma^{-1}})^\wp = \xi^\wp = 1$, so that $a = \alpha^\wp \in A_K$; then $\alpha \in \wp^{-1}(\Delta)$, and therefore $M \subseteq K(\wp^{-1}(\Delta))$. $\square$

As the main result of general Kummer theory, we now obtain the following

(3.3) Theorem. *The correspondence*

$$\Delta \longmapsto L = K(\wp^{-1}(\Delta))$$

is a 1-1-correspondence between the groups Δ such that $A_K^\wp \subseteq \Delta \subseteq A_K$ and the abelian extensions $L|K$ of exponent n .

If Δ and L correspond to each other, then $A_L^\wp \cap A_K = \Delta$, and we have a canonical isomorphism

$$\Delta/A_K^\wp \cong \text{Hom}(G(L|K), \mu_\wp), \quad a \bmod A_K^\wp \longmapsto \chi_a,$$

where the character $\chi_a : G(L|K) \rightarrow \mu_\wp$ is given by $\chi_a(\sigma) = \alpha^{\sigma^{-1}}$, for $\alpha \in \wp^{-1}(a)$.

Proof: Let $L|K$ be an abelian extension of exponent n . By (3.2), we then find $L = K(\wp^{-1}(\Delta))$ with $\Delta = A_L^\wp \cap A_K$. We consider the homomorphism

$$\Delta \longrightarrow \text{Hom}(G(L|K), \mu_\wp), \quad a \longmapsto \chi_a,$$

where $\chi_a(\sigma) = \alpha^{\sigma^{-1}}$, $\alpha \in \wp^{-1}(a)$. Since

$$\begin{aligned} \chi_a = 1 &\iff \alpha^{\sigma^{-1}} = 1 \text{ for all } \sigma \in G(L|K) \\ &\iff \alpha \in A_K \iff a = \alpha^\wp \in A_K^\wp, \end{aligned}$$

it has the kernel $A_K^\wp$. To prove the surjectivity, we let $\chi \in \text{Hom}(G(L|K), \mu_\wp)$. χ defines a cyclic extension $M|K$ and is the composite of homomorphisms $G(L|K) \rightarrow G(M|K) \xrightarrow{\bar{\chi}} \mu_\wp$. Let σ be a generator of $G(M|K)$. Since

$N_{M|K}(\bar{\chi}(\sigma)) = \bar{\chi}(\sigma)^{[M:K]} = 1$, we deduce from (3.1) that $\bar{\chi}(\sigma) = \alpha^{\sigma-1}$ for some $\alpha \in A_M$. Now, $(\alpha^{\wp})^{\sigma-1} = (\alpha^{\sigma-1})^{\wp} = \bar{\chi}(\sigma)^{\wp} = 1$, so that $a = \alpha^{\wp} \in A_L^{\wp} \cap A_K = \Delta$. For $\tau \in G(L|K)$, one has $\chi(\tau) = \bar{\chi}(\tau|_M) = \alpha^{\tau-1} = \chi_a(\tau)$, so that $\chi = \chi_a$. This proves the surjectivity, and we obtain an isomorphism

$$\Delta/A_K^{\wp} \cong \text{Hom}(G(L|K), \mu_{\wp}).$$

If Δ is any group between $A_K^{\wp}$ and A_K and if $L = K(\wp^{-1}(\Delta))$, then $\Delta = A_L^{\wp} \cap A_K$. In fact, putting $\Delta' = A_L^{\wp} \cap A_K$, we have just seen that one has

$$\Delta'/A_K^{\wp} \cong \text{Hom}(G(L|K), \mu_{\wp}).$$

The subgroup $\Delta/A_K^{\wp}$ corresponds under Pontryagin duality to the subgroup $\text{Hom}(G(L|K)/H, \mu_{\wp})$, where

$$H = \{ \sigma \in G(L|K) \mid \chi_a(\sigma) = 1 \text{ for all } a \in \Delta \}.$$

As $\alpha^{\sigma-1} = \chi_a(\sigma)$ for $\alpha \in \wp^{-1}(a)$, H leaves fixed the elements of $\wp^{-1}(\Delta)$, and as $K(\wp^{-1}(\Delta)) = L$, we find that $H = 1$, so that $\text{Hom}(G(L|K)/H, \mu_{\wp}) = \text{Hom}(G(L|K), \mu_{\wp})$. It follows that $\Delta/A_K^{\wp} = \Delta'/A_K^{\wp}$, i.e., $\Delta = \Delta'$.

It is therefore clear that the correspondence $\Delta \mapsto L = K(\wp^{-1}(\Delta))$ is a 1-1-correspondence, as claimed. This finishes the proof of the theorem. $\square$

Remarks and Examples: 1) If $L|K$ is infinite, then $\text{Hom}(G(L|K), \mu_{\wp})$ has to be interpreted as the group of all *continuous* homomorphisms $\chi : G(L|K) \rightarrow \mu_{\wp}$, i.e., as the character group of the *topological* group $G(L|K)$.

2) The composite of two abelian extensions of K of exponent n is again of the same type, and all of them lie in the *maximal* abelian extension of exponent n . It is given by $\hat{K} = K(\wp^{-1}(A_K))$, and for the Pontryagin dual

$$G(\hat{K}|K)^* = \text{Hom}(G(\hat{K}|K), \mathbb{Q}/\mathbb{Z}) = \text{Hom}(G(\hat{K}|K), \mu_{\wp})$$

we have by (3.3) that

$$G(\hat{K}|K)^* \cong A_K/A_K^{\wp}.$$

3) If k is an actual field of positive characteristic p and $\bar{k}$ is the separable closure of k , then A may be chosen to be the additive group $\bar{k}$ and $\wp$ to be the operator

$$\wp : \bar{k} \longrightarrow \bar{k}, \quad a \longmapsto \wp a = a^p - a.$$

Then axiom (3.1) is indeed satisfied, for we have, in complete generality:

(3.4) Proposition. *For every cyclic finite field extension $L|K$, one has $H^{-1}(G(L|K), L) = 1$.*

Proof: The extension $L|K$ always admits a normal basis $\{\sigma c \mid \sigma \in G(L|K)\}$, so that $L = \bigoplus_{\sigma} K \sigma c$. This means that L is a $G(L|K)$ -induced module in the sense of § 7, and then $H^{-1}(G(L|K), L) = 1$, by (7.4). $\square$

The Kummer theory with respect to the operator $\wp a = a^p - a$ is usually called **Artin-Schreier theory**.

4) The chief application of the theory developed above is to the case where G is the absolute Galois group $G(\bar{k}|k)$ of an actual field k , A is the multiplicative group $\bar{k}^*$ of the algebraic closure, and $\wp$ is the n -th power map $a \mapsto a^n$, for some natural number n which is relatively prime to the characteristic of k (in particular, n is arbitrary if $\text{char}(k) = 0$). Axiom (3.1) is always satisfied in this case and is called **Hilbert 90** because this statement occurs as *Satz* number 90 among the 169 theorems in Hilbert's famous "Zahlbericht" [72]. Thus we have the

(3.5) Theorem (Hilbert 90). *For a cyclic field extension $L|K$ one always has*

$$H^{-1}(G(L|K), L^*) = 1.$$

In other words:

An element $\alpha \in L^$ of norm $N_{L|K}(\alpha) = 1$ is of the form $\alpha = \beta^{\sigma^{-1}}$, where $\beta \in L^*$ and σ is a generator of $G(L|K)$.*

Proof: Let $n = [L : K]$. By virtue of the linear independence of the automorphisms $1, \sigma, \dots, \sigma^{n-1}$ (see [15], chap. 5, § 7, no. 5), there exists an element $\gamma \in L^*$ such that

$$\beta = \gamma + \alpha \gamma^{\sigma} + \alpha^{1+\sigma} \gamma^{\sigma^2} + \dots + \alpha^{1+\sigma+\dots+\sigma^{n-2}} \gamma^{\sigma^{n-1}} \neq 0.$$

As $N_{L|K}(\alpha) = 1$, one gets $\alpha \beta^{\sigma} = \beta$, and thus $\alpha = \beta^{1-\sigma}$. $\square$

If now the field K contains the group μ_n of n -th roots of unity, the operator $\wp(a) = a^n$ has exponent n , and we obtain the following corollary, which is the most important special case of theorem (3.3).

(3.6) Corollary. *Let n be a natural number which is relatively prime to the characteristic of the field K , and assume that $\mu_n \subseteq K$.*

Then the abelian extensions $L|K$ of exponent n correspond 1–1 to the subgroups $\Delta \subseteq K^$ which contain K^{*n} , via the rule*

$$\Delta \mapsto L = K(\sqrt[n]{\Delta}),$$

and we have

$$G(L|K) \cong \text{Hom}(\Delta/K^{*n}, \mu_n).$$

Hilbert's theorem 90, which is the main basis of this corollary, admits the following generalization to arbitrary Galois extensions $L|K$, which goes back to the mathematician *EMMY NOETHER* (1882–1935). Let G be a finite group and A a multiplicative G -module. A **1-cocycle**, or **crossed homomorphism**, of G with values in A is a function $f : G \rightarrow A$ satisfying

$$f(\sigma\tau) = f(\sigma)^\tau f(\tau)$$

for all $\sigma, \tau \in G$. The 1-cocycles form an abelian group $Z^1(G, A)$. For every $a \in A$, the function

$$f_a(\sigma) = a^{\sigma-1}$$

is a 1-cocycle, for one has

$$f_a(\sigma\tau) = a^{\sigma\tau-1} = (a^{\sigma-1})^\tau a^{\tau-1} = f_a(\sigma)^\tau f_a(\tau).$$

The functions f_a are called **1-coboundaries** and form a subgroup $B^1(G, A)$ of $Z^1(G, A)$. We define

$$H^1(G, A) = Z^1(G, A)/B^1(G, A)$$

and obtain as a first result about this group the

(3.7) Proposition. *If G is cyclic, then $H^1(G, A) \cong H^{-1}(G, A)$.*

Proof: Let $G = \langle \sigma \rangle$. If $f \in Z^1(G, A)$, then for $k \geq 1$

$$f(\sigma^k) = f(\sigma^{k-1})^\sigma f(\sigma) = f(\sigma^{k-2})^{\sigma^2} f(\sigma)^\sigma f(\sigma) = \cdots = \prod_{i=0}^{k-1} f(\sigma)^{\sigma^i},$$

and $f(1) = 1$ because $f(1) = f(1)f(1)$. If $n = \#G$, then

$$N_G f(\sigma) = \prod_{i=0}^{n-1} f(\sigma)^{\sigma^i} = f(\sigma^n) = f(1) = 1,$$

so that $f(\sigma) \in {}_{N_G}A = \{a \in A \mid N_G a = \prod_{i=0}^{n-1} a^{\sigma^i} = 1\}$. Conversely we obtain, for every $a \in A$ such that $N_G a = 1$, a 1-cocycle by putting $f(\sigma) = a$ and

$$f(\sigma^k) = \prod_{i=0}^{k-1} a^{\sigma^i}.$$

The reader is invited to check this. The map $f \mapsto f(\sigma)$ therefore is an isomorphism between $Z^1(G, A)$ and ${}_{N_G}A$. This isomorphism maps $B^1(G, A)$ onto $I_G A$, because $f \in B^1(G, A) \iff f(\sigma^k) = a^{\sigma^{k-1}}$ for some fixed $a \iff f(\sigma) = a^{\sigma^{-1}} \iff f(\sigma) \in I_G A$. $\square$

Noether's generalization of Hilbert's theorem 90 now reads:

(3.8) Proposition. *For a finite Galois field extension $L|K$, one has that*

$$H^1(G(L|K), L^*) = 1.$$

Proof: Let $f : G \rightarrow L^*$ be a 1-cocycle. For $c \in L^*$, we put

$$\alpha = \sum_{\sigma \in G(L|K)} f(\sigma) c^{\sigma}.$$

Since the automorphisms σ are linearly independent (see [15], chap. 5, §7, no. 5), we can choose $c \in L^*$ such that $\alpha \neq 0$. For $\tau \in G(L|K)$, we obtain

$$\alpha^{\tau} = \sum_{\sigma} f(\sigma)^{\tau} c^{\sigma^{\tau}} = \sum_{\sigma} f(\tau)^{-1} f(\sigma\tau) c^{\sigma^{\tau}} = f(\tau)^{-1} \alpha,$$

i.e., $f(\tau) = \beta^{\tau-1}$ with $\beta = \alpha^{-1}$. $\square$

This proposition will only be applied once in this book (see chap. VI, (2.5)).

Exercise 1. Show that Hilbert 90 in Noether's formulation also holds for the additive group L of a Galois extension $L|K$.

Hint: Use the normal basis theorem.

Exercise 2. Let k be a field of characteristic p and $\bar{k}$ its separable closure. For fixed $n \geq 1$, consider in the ring of Witt vectors $W(\bar{k})$ (see chap. II, §4, exercise 2–6) the additive group $W_n(\bar{k})$ of truncated Witt vectors $a = (a_0, a_1, \dots, a_{n-1})$. Show that axiom (3.1) holds for the $G(\bar{k}|k)$ -module $A = W_n(\bar{k})$.

Exercise 3. Show that the operator

$$\wp : W_n(\bar{k}) \rightarrow W_n(\bar{k}), \quad \wp a = Fa - a,$$

is a homomorphism with cyclic kernel $\mu_{\wp}$ of order p^n . Discuss the corresponding Kummer theory for the abelian extensions of exponent p^n .

Exercise 4. Let G be a profinite group and A a continuous G -module. Put

$$H^1(G, A) = Z^1(G, A)/B^1(G, A),$$

where $Z^1(G, A)$ consists of all continuous maps $f : G \rightarrow A$ (with respect to the discrete topology on A) such that $f(\sigma\tau) = f(\sigma)^\tau f(\tau)$, and $B^1(G, A)$ consists of all functions of the form $f_a(\sigma) = a^{\sigma-1}$, $a \in A$. Show that if g is a closed normal subgroup of G , then one has an exact sequence

$$1 \rightarrow H^1(G/g, A^g) \rightarrow H^1(G, A) \rightarrow H^1(g, A).$$

Exercise 5. Show that $H^1(G, A) = \varinjlim H^1(G/N, A^N)$, where N varies over all the open normal subgroups of G .

Exercise 6. If $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ is an exact sequence of continuous G -modules, then one has an exact sequence

$$1 \rightarrow A^G \rightarrow B^G \rightarrow C^G \rightarrow H^1(G, A) \rightarrow H^1(G, B) \rightarrow H^1(G, C).$$

Remark: The group $H^1(G, A)$ is only the first term of a whole series of groups $H^i(G, A)$, $i = 1, 2, 3, \dots$, which are the objects of **group cohomology** (see [145]). Class field theory can also be built upon this theory (see [10], [108]).

Exercise 7. Even for infinite Galois extensions $L|K$, one has Hilbert's theorem 90: $H^1(G(L|K), L^*) = 1$.

Exercise 8. If n is not divisible by the characteristic of the field K and if μ_n denotes the group of n -th roots of unity in the separable closure $\bar{K}$, then

$$H^1(G_K, \mu_n) \cong K^*/K^{*n}.$$

§ 4. Abstract Valuation Theory

The further development will now be based on a fixed choice of a surjective continuous homomorphism

$$d : G \longrightarrow \widehat{\mathbb{Z}}$$

from the profinite group G onto the procyclic group $\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z}$ (see § 2, example 4). This homomorphism will produce a theory which is an abstract reflection of the ramification theory of p -adic number fields. Indeed, in the case where G is the absolute Galois group $G_k = G(\bar{k}|k)$ of a p -adic number field k , such a surjective homomorphism $d : G \rightarrow \widehat{\mathbb{Z}}$ arises via the maximal unramified extension $\bar{k}|k$: if $\mathbb{F}_q$ is the residue class field of k , then, by chap. II, § 9, p. 173 and example 5 in § 2, we have canonical isomorphisms

$$G(\bar{k}|k) \cong G(\bar{\mathbb{F}}_q|\mathbb{F}_q) \cong \widehat{\mathbb{Z}}$$

which associate to the element $1 \in \widehat{\mathbb{Z}}$ the Frobenius automorphism $\varphi \in G(\tilde{k}|k)$. It is defined by

$$a^\varphi \equiv a^q \pmod{\tilde{\mathfrak{p}}} \quad \text{for } a \in \tilde{\mathcal{O}},$$

where $\tilde{\mathcal{O}}$, resp. $\tilde{\mathfrak{p}}$, denote the valuation ring of $\tilde{k}$, resp. its maximal ideal. The homomorphism $d : G \rightarrow \widehat{\mathbb{Z}}$ in question is then given, in this concrete case, as the composite

$$G \longrightarrow G(\tilde{k}|k) \xrightarrow{\sim} \widehat{\mathbb{Z}}.$$

In the abstract situation, the initial choice of a surjective homomorphism $d : G \rightarrow \widehat{\mathbb{Z}}$ mimics the p -adic case, but the applications of the theory are by no means confined to p -adic number fields. The kernel I of d has a certain fixed field $\tilde{k}|k$, and d induces an isomorphism $G(\tilde{k}|k) \cong \widehat{\mathbb{Z}}$.

More generally, for any field K we denote by I_K the kernel of the restriction $d : G_K \rightarrow \widehat{\mathbb{Z}}$, and call it the **inertia group** over K . Since

$$I_K = G_K \cap I = G_K \cap G_{\tilde{k}} = G_{K\tilde{k}},$$

the fixed field $\tilde{K}$ of I_K is the composite

$$\tilde{K} = K\tilde{k}.$$

We call $\tilde{K}|K$ the **maximal unramified extension** of K . We put

$$f_K = (\widehat{\mathbb{Z}} : d(G_K)), \quad e_K = (I : I_K)$$

and obtain, when f_K is finite, a surjective homomorphism

$$d_K = \frac{1}{f_K} d : G_K \longrightarrow \widehat{\mathbb{Z}}$$

with kernel I_K , and an isomorphism

$$d_K : G(\tilde{K}|K) \xrightarrow{\sim} \widehat{\mathbb{Z}}.$$

(4.1) Definition. The element $\varphi_K \in G(\tilde{K}|K)$ such that $d_K(\varphi_K) = 1$ is called the **Frobenius** over K .

For a field extension $L|K$ we define the **inertia degree** $f_{L|K}$ and the **ramification index** $e_{L|K}$ by

$$f_{L|K} = (d(G_K) : d(G_L)) \quad \text{and} \quad e_{L|K} = (I_K : I_L).$$

For a tower of fields $K \subseteq L \subseteq M$ this definition obviously implies that

$$f_{M|K} = f_{L|K} f_{M|L} \quad \text{and} \quad e_{M|K} = e_{L|K} e_{M|L}.$$

(4.2) Proposition. *For every extension $L|K$ we have the “fundamental identity”*

$$[L : K] = f_{L|K} e_{L|K}.$$

Proof: The exact commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & I_L & \longrightarrow & G_L & \longrightarrow & d(G_L) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & I_K & \longrightarrow & G_K & \longrightarrow & d(G_K) \longrightarrow 1 \end{array}$$

immediately yields, if $L|K$ is Galois, the exact sequence

$$1 \longrightarrow I_K/I_L \longrightarrow G(L|K) \longrightarrow d(G_K)/d(G_L) \longrightarrow 1.$$

If $L|K$ is not Galois, we pass to a Galois extension $M|K$ containing L , and get the result from the above transitivity rules for e and f . $\square$

$L|K$ is called **unramified** if $e_{L|K} = 1$, i.e., if $L \subseteq \tilde{K}$. $L|K$ is called **totally ramified** if $f_{L|K} = 1$, i.e., if $L \cap \tilde{K} = K$. In the unramified case, we have the surjective homomorphism

$$G(\tilde{K}|K) \rightarrow G(L|K)$$

and, if $f_K < \infty$, we call the image $\varphi_{L|K}$ of φ_K the **Frobenius automorphism** of $L|K$.

For an arbitrary extension $L|K$ one has

$$\tilde{L} = L\tilde{K},$$

since $L\tilde{K} = LK\tilde{k} = L\tilde{k} = \tilde{L}$, and $L \cap \tilde{K}|K$ is the maximal unramified subextension of $L|K$. It clearly has degree

$$f_{L|K} = [L \cap \tilde{K} : K].$$

Equally obvious is the

(4.3) Proposition. *If f_K and f_L are finite, then $f_{L|K} = f_L/f_K$, and we have the commutative diagram*

$$\begin{array}{ccc} G_L & \xrightarrow{d_L} & \widehat{\mathbb{Z}} \\ \downarrow & & \downarrow f_{L|K} \\ G_K & \xrightarrow{d_K} & \widehat{\mathbb{Z}}. \end{array}$$

In particular, one has $\varphi_L|_{\tilde{K}} = \varphi_K^{f_{L|K}}$.

The Frobenius automorphism governs the entire class field theory like a king. It is therefore most remarkable that in the case of a finite Galois extension $L|K$, every $\sigma \in G(L|K)$ becomes a Frobenius automorphism once it is manœuvred into the right position. This is achieved in the following manner. For what follows, let us assume systematically that $f_K < \infty$. We pass from the Galois extension $L|K$ to the extension $\tilde{L}|K$ and consider in the Galois group $G(\tilde{L}|K)$ the semigroup

$$\text{Frob}(\tilde{L}|K) = \{ \sigma \in G(\tilde{L}|K) \mid d_K(\sigma) \in \mathbb{N} \}.$$

Observe here that $d_K : G_K \longrightarrow \widehat{\mathbb{Z}}$ factorizes through $G(\tilde{L}|K)$ because $G_{\tilde{L}} = I_L \subseteq I_K$; recall also that $0 \notin \mathbb{N}$. Firstly, we have the

(4.4) Proposition. *For a finite Galois extension $L|K$ the mapping*

$$\text{Frob}(\tilde{L}|K) \longrightarrow G(L|K), \quad \sigma \longmapsto \sigma|_L,$$

is surjective.

Proof: Let $\sigma \in G(L|K)$ and let $\varphi \in G(\tilde{L}|K)$ be an element such that $d_K(\varphi) = 1$. Then $\varphi|_{\tilde{K}} = \varphi_K$ and $\varphi|_{L \cap \tilde{K}} = \varphi_{L \cap \tilde{K}|K}$. Restricting σ to the maximal unramified subextension $L \cap \tilde{K}|K$, it becomes a power of the Frobenius automorphism, $\sigma|_{L \cap \tilde{K}} = \varphi^n|_{L \cap \tilde{K}}$, so we may choose n in $\mathbb{N}$. As $\tilde{L} = L\tilde{K}$, we have

$$G(\tilde{L}|\tilde{K}) \cong G(L|L \cap \tilde{K}).$$

If now $\tau \in G(\tilde{L}|\tilde{K})$ is mapped to $\sigma\varphi^{-n}|_L$ under this isomorphism, then $\tilde{\sigma} = \tau\varphi^n$ is an element satisfying $\tilde{\sigma}|_L = \tau\varphi^n|_L = \sigma\varphi^{-n}\varphi^n|_L = \sigma$ and $\tilde{\sigma}|_{\tilde{K}} = \varphi_K^n$. Hence $d_K(\tilde{\sigma}) = n$, and so $\tilde{\sigma} \in \text{Frob}(\tilde{L}|K)$. $\square$

Thus every element $\sigma \in G(L|K)$ may be lifted to an element $\tilde{\sigma} \in \text{Frob}(\tilde{L}|K)$. The following proposition shows that this lifting, considered over its fixed field, is actually the Frobenius automorphism.

(4.5) Proposition. *Let $\tilde{\sigma} \in \text{Frob}(\tilde{L}|K)$, and let Σ be the fixed field of $\tilde{\sigma}$. Then we have:*

$$(i) \quad f_{\Sigma|K} = d_K(\tilde{\sigma}), \quad (ii) \quad [\Sigma : K] < \infty, \quad (iii) \quad \tilde{\Sigma} = \tilde{L}, \quad (iv) \quad \tilde{\sigma} = \varphi_{\Sigma}.$$

Proof: (i) $\Sigma \cap \tilde{K}$ is the fixed field of $\tilde{\sigma}|_{\tilde{K}} = \varphi_K^{d_K(\tilde{\sigma})}$, so that

$$f_{\Sigma|K} = [\Sigma \cap \tilde{K} : K] = d_K(\tilde{\sigma}).$$

(ii) One has $\tilde{K} \subseteq \Sigma\tilde{K} = \tilde{\Sigma} \subseteq \tilde{L}$; thus

$$e_{\Sigma|K} = (I_K : I_{\Sigma}) = \#G(\tilde{\Sigma}|\tilde{K}) \leq \#G(\tilde{L}|\tilde{K})$$

is finite. Therefore $[\Sigma : K] = f_{\Sigma|K}e_{\Sigma|K}$ is finite as well.

(iii) The canonical surjection $\Gamma = G(\tilde{L}|\Sigma) \rightarrow G(\tilde{\Sigma}|\Sigma) \cong \widehat{\mathbb{Z}}$ has to be bijective. For since $\Gamma = \langle \tilde{\sigma} \rangle$ is procyclic, one finds $(\Gamma : \Gamma^n) \leq n$ for every $n \in \mathbb{N}$ (see §2, p. 273). Thus the induced maps $\Gamma/\Gamma^n \cong \widehat{\mathbb{Z}}/n\widehat{\mathbb{Z}}$ are bijective and so is $\Gamma \rightarrow \widehat{\mathbb{Z}}$. But $G(\tilde{L}|\Sigma) = G(\tilde{\Sigma}|\Sigma)$ implies that $\tilde{L} = \tilde{\Sigma}$.

(iv) $f_{\Sigma|K}d_{\Sigma}(\tilde{\sigma}) = d_K(\tilde{\sigma}) = f_{\Sigma|K}$; thus $d_{\Sigma}(\tilde{\sigma}) = 1$, and so $\tilde{\sigma} = \varphi_{\Sigma}$. $\square$

Let us illustrate the situation described in the last proposition by a diagram, which one should keep in mind for the sequel.

$$\begin{array}{ccc}
 & L & \xrightarrow{\varphi_L} \tilde{L} = \tilde{\Sigma} \\
 & \downarrow & \swarrow \tilde{\sigma} = \varphi_{\Sigma} \\
 & \Sigma & \\
 f_{L|K} \nearrow & \downarrow & \downarrow \\
 K & & \tilde{K} \\
 & \xleftarrow{f_{\Sigma|K}} & \xleftarrow{\varphi_K}
 \end{array}$$

All the preceding discussions arose entirely from the initial datum of the homomorphism $d : G \rightarrow \widehat{\mathbb{Z}}$. We now add to the data a multiplicative G -module A , which we equip with a homomorphism that is to play the rôle of a henselian valuation.

(4.6) Definition. A **henselian valuation** of A_k with respect to $d : G \rightarrow \widehat{\mathbb{Z}}$ is a homomorphism

$$v : A_k \rightarrow \widehat{\mathbb{Z}}$$

satisfying the following properties:

- (i) $v(A_k) = Z \supseteq \mathbb{Z}$ and $Z/nZ \cong \mathbb{Z}/n\mathbb{Z}$ for all $n \in \mathbb{N}$,
- (ii) $v(N_{K|k}A_K) = f_K Z$ for all finite extensions $K|k$.

Exactly like the original homomorphism $d : G_k \rightarrow \widehat{\mathbb{Z}}$, the henselian valuation $v : A_k \rightarrow \widehat{\mathbb{Z}}$ has the property of reproducing itself over every finite extension K of k .

(4.7) Proposition. *For every field K which is finite over k , the formula*

$$v_K = \frac{1}{f_K} v \circ N_{K|k} : A_K \longrightarrow \mathbb{Z}$$

defines a surjective homomorphism satisfying the following properties:

- (i) $v_K = v_{K^\sigma} \circ \sigma$ for all $\sigma \in G$.
- (ii) For every finite extension $L|K$, one has the commutative diagram

$$\begin{array}{ccc} A_L & \xrightarrow{v_L} & \widehat{\mathbb{Z}} \\ N_{L|K} \downarrow & & \downarrow f_{L|K} \\ A_K & \xrightarrow{v_K} & \widehat{\mathbb{Z}}. \end{array}$$

Proof: (i) If τ runs through a system of representatives of G_k/G_K , then $\sigma^{-1}\tau\sigma$ sweeps across a system of representatives of $G_k/\sigma^{-1}G_K\sigma = G_k/G_{K^\sigma}$. Hence we have, for $a \in A_K$,

$$\begin{aligned} v_{K^\sigma}(a^\sigma) &= \frac{1}{f_{K^\sigma}} v\left(\prod_{\tau} a^{\sigma\sigma^{-1}\tau\sigma}\right) = \frac{1}{f_K} v\left(\left(\prod_{\tau} a^{\tau}\right)^{\sigma}\right) = \frac{1}{f_K} v(N_{K|k}(a)) \\ &= v_K(a). \end{aligned}$$

(ii) For $a \in A_L$ one has:

$$\begin{aligned} f_{L|K} v_L(a) &= f_{L|K} \frac{1}{f_L} v(N_{L|k}(a)) = \frac{1}{f_K} v(N_{K|k}(N_{L|K}(a))) \\ &= v_K(N_{L|K}(a)). \end{aligned} \quad \square$$

(4.8) Definition. A **prime element** of A_K is an element $\pi_K \in A_K$ such that $v_K(\pi_K) = 1$. We put

$$U_K = \{u \in A_K \mid v_K(u) = 0\}.$$

For an unramified extension $L|K$, that is, an extension such that $f_{L|K} = [L : K]$, we have from (4.7), (ii) that $v_L|_{A_K} = v_K$. In particular, a prime element of A_K is itself also a prime element of A_L . If on the other hand, $L|K$ is totally ramified, i.e., $f_{L|K} = 1$, and if π_L is a prime element of A_L , then $\pi_K = N_{L|K}(\pi_L)$ is a prime element of A_K .

Exercise 1. Assume that every closed abelian subgroup of G is procyclic. Let $K|k$ be a finite extension. A **microprime** $\mathfrak{p}$ of K is by definition a conjugacy class $\langle \sigma \rangle \subseteq G_K$ of some Frobenius element $\sigma \in \text{Frob}(\bar{k}|K)$ which is not a proper power σ'^n , $n > 1$, of some other Frobenius element $\sigma' \in \text{Frob}(\bar{k}|K)$. Let $\text{spec}(K)$ be the set of all microprimes of K . Show that if $L|K$ is a finite extension, then there is a canonical mapping

$$\pi : \text{spec}(L) \rightarrow \text{spec}(K).$$

Above any microprime $\mathfrak{p}$ there are only finitely many microprimes $\mathfrak{P}$ of L , i.e., the set $\pi^{-1}(\mathfrak{p})$ is finite. We write $\mathfrak{P}|\mathfrak{p}$ to mean $\mathfrak{P} \in \pi^{-1}(\mathfrak{p})$.

Exercise 2. For a finite extension $L|K$ and a microprime $\mathfrak{P}|\mathfrak{p}$ of L , let $f_{\mathfrak{P}|\mathfrak{p}} = d(\mathfrak{P})/d(\mathfrak{p})$. Show that

$$\sum_{\mathfrak{P}|\mathfrak{p}} f_{\mathfrak{P}|\mathfrak{p}} = [L : K].$$

Exercise 3. For an infinite extension $L|K$, let

$$\text{spec}(L) = \varprojlim_{\alpha} \text{spec}(L_{\alpha}),$$

where $L_{\alpha}|K$ varies over the finite subextensions of $L|K$. What are the microprimes of $\bar{k}$?

Exercise 4. Show that if $L|K$ is Galois, then the Galois group $G(L|K)$ operates transitively on $\text{spec}(L)$. The “decomposition group”

$$G_{\mathfrak{P}}(L|K) = \{\sigma \in G(L|K) \mid \mathfrak{P}^{\sigma} = \mathfrak{P}\}$$

is cyclic, and if $Z_{\mathfrak{P}} = L^{G_{\mathfrak{P}}(L|K)}$ is the “decomposition field” of $\mathfrak{P} \in \text{spec}(L)$, then $L|Z_{\mathfrak{P}}$ is unramified.

§ 5. The Reciprocity Map

Continuing with the notation of the previous section, we consider again a profinite group G , a continuous G -module A , and a pair of homomorphisms

$$d : G \longrightarrow \widehat{\mathbb{Z}}, \quad v : A_k \longrightarrow \widehat{\mathbb{Z}},$$

such that d is continuous and surjective and v is a henselian valuation with respect to d . In the following we introduce the convention that the letter K , whenever it occurs without embellishments or commentary to the contrary, will always denote a field of *finite* degree over k . We furthermore impose the following axiomatic condition, which will be systematically assumed in the sequel.

(5.1) Axiom. For every unramified finite extension $L|K$ one has

$$H^i(G(L|K), U_L) = 1 \quad \text{for } i = 0, -1.$$

For an infinite extension $L|K$ we set

$$N_{L|K} A_L = \bigcap_M N_{M|K} A_M,$$

with $M|K$ varying over the finite subextensions of $L|K$.

Our goal is to define a canonical homomorphism

$$r_{L|K} : G(L|K) \longrightarrow A_K / N_{L|K} A_L$$

for every finite Galois extension $L|K$. To this end, we pass from $L|K$ to the extension $\tilde{L}|K$ and define first a mapping on the semigroup

$$\text{Frob}(\tilde{L}|K) = \{ \sigma \in G(\tilde{L}|K) \mid d_K(\sigma) \in \mathbb{N} \}.$$

(5.2) Definition. *The reciprocity map*

$$r_{\tilde{L}|K} : \text{Frob}(\tilde{L}|K) \longrightarrow A_K / N_{\tilde{L}|K} A_{\tilde{L}}$$

is defined by

$$r_{\tilde{L}|K}(\sigma) = N_{\Sigma|K}(\pi_\Sigma) \bmod N_{\tilde{L}|K} A_{\tilde{L}},$$

where Σ is the fixed field of σ and $\pi_\Sigma \in A_\Sigma$ is a prime element.

Observe that Σ is of finite degree over K by (4.5), and σ becomes the Frobenius automorphism φ_Σ over Σ . The definition of $r_{\tilde{L}|K}(\sigma)$ does not depend on the choice of the element π_Σ . For another one differs from π_Σ only by an element $u \in U_\Sigma$, and for this we have $N_{\Sigma|K}(u) \in N_{\tilde{L}|K} A_{\tilde{L}}$, so that $N_{\Sigma|K}(u) \in N_{M|K} A_M$ for every finite Galois subextension $M|K$ of $\tilde{L}|K$. To see this, we may clearly assume that $\Sigma \subseteq M$. Applying (5.1) to the unramified extension $M|\Sigma$, one finds $u = N_{M|\Sigma}(\varepsilon)$, $\varepsilon \in U_M$, and thus

$$N_{\Sigma|K}(u) = N_{\Sigma|K}(N_{M|\Sigma}(\varepsilon)) = N_{M|K}(\varepsilon) \in N_{M|K} A_M.$$

Next we want to show that the reciprocity map $r_{\tilde{L}|K}$ is multiplicative. To do this, we consider for every $\sigma \in G(\tilde{L}|K)$ and every $n \in \mathbb{N}$ the endomorphisms

$$\begin{aligned} \sigma - 1 : A_{\tilde{L}} &\longrightarrow A_{\tilde{L}}, & a &\longmapsto a^{\sigma-1} = a^\sigma / a, \\ \sigma_n : A_{\tilde{L}} &\longrightarrow A_{\tilde{L}}, & a &\longmapsto a^{\sigma_n} = \prod_{i=0}^{n-1} a^{\sigma^i}. \end{aligned}$$

In formal notation, this gives $\sigma_n = \frac{\sigma^n - 1}{\sigma - 1}$, and we find that

$$(\sigma - 1) \circ \sigma_n = \sigma_n \circ (\sigma - 1) = \sigma^n - 1.$$

Now we introduce the homomorphism

$$N = N_{\tilde{L}|\tilde{K}} : A_{\tilde{L}} \longrightarrow A_{\tilde{K}}$$

and prove two lemmas for it.

(5.3) Lemma. *Let $\varphi, \sigma \in \text{Frob}(\tilde{L}|K)$ with $d_K(\varphi) = 1$, $d_K(\sigma) = n$. If Σ is the fixed field of σ and $a \in A_\Sigma$, then*

$$N_{\Sigma|K}(a) = (N \circ \varphi_n)(a) = (\varphi_n \circ N)(a).$$

Proof: The maximal unramified subextension $\Sigma^0 = \Sigma \cap \tilde{K}|K$ is of degree n , and its Galois group $G(\Sigma^0|K)$ is generated by the Frobenius automorphism $\varphi_{\Sigma^0|K} = \varphi_K|_{\Sigma^0} = \varphi|_{\tilde{K}}|_{\Sigma^0} = \varphi|_{\Sigma^0}$. Consequently, $N_{\Sigma^0|K} = \varphi_n|_{A_{\Sigma^0}}$. On the other hand, one has $\Sigma\tilde{K} = \tilde{L}$ and $\Sigma \cap \tilde{K} = \Sigma^0$, and therefore $N_{\Sigma|\Sigma^0} = N|_{A_\Sigma}$. For $a \in A_\Sigma$ we thus get

$$N_{\Sigma|K}(a) = N_{\Sigma^0|K}(N_{\Sigma|\Sigma^0}(a)) = N(a)^{\varphi_n} = N(a^{\varphi_n}).$$

The last equation follows from $\varphi G(\tilde{L}|\tilde{K}) = G(\tilde{L}|\tilde{K})\varphi$. $\square$

The subgroup $I_{G(\tilde{L}|\tilde{K})}U_{\tilde{L}}$, which is generated by all elements of the form $u^{\tau-1}$, $u \in U_{\tilde{L}}$, $\tau \in G(\tilde{L}|\tilde{K})$, is mapped to 1 by the homomorphism $N = N_{\tilde{L}|\tilde{K}} : U_{\tilde{L}} \rightarrow U_{\tilde{K}}$. We therefore obtain an induced homomorphism

$$N : H_0(G(\tilde{L}|\tilde{K}), U_{\tilde{L}}) \rightarrow U_{\tilde{K}}$$

on the quotient group $H_0(G(\tilde{L}|\tilde{K}), U_{\tilde{L}}) = U_{\tilde{L}}/I_{G(\tilde{L}|\tilde{K})}U_{\tilde{L}}$. For this group, we have the following lemma.

(5.4) Lemma. *If $x \in H_0(G(\tilde{L}|\tilde{K}), U_{\tilde{L}})$ is fixed by an element $\varphi \in G(\tilde{L}|K)$ such that $d_K(\varphi) = 1$, i.e., $x^\varphi = x$, then*

$$N(x) \in N_{\tilde{L}|K}U_{\tilde{L}}.$$

Proof: Let $x = u \bmod I_{G(\tilde{L}|\tilde{K})}U_{\tilde{L}}$, with $x^{\varphi-1} = 1$, so that

$$(*) \quad u^{\varphi-1} = \prod_{i=1}^r u_i^{\tau_i-1}, \quad u_i \in U_{\tilde{L}}, \quad \tau_i \in G(\tilde{L}|\tilde{K}).$$

Let $M|K$ be a finite Galois subextension of $\tilde{L}|K$. In order to prove that $N(u) \in N_{M|K}U_M$, we may assume that $u, u_i \in U_M$ and $L \subseteq M$. Let $n = [M : K]$, $\sigma = \varphi^n$ and let $\Sigma \supseteq M$ be the fixed field of σ . Further, let $\Sigma_n|\Sigma$ be the unramified extension of degree n , i.e., the fixed field of $\sigma^n = \varphi_n$. By (5.1), we can then find elements $\tilde{u}, \tilde{u}_i \in U_{\Sigma_n}$ such that

$$u = N_{\Sigma_n|\Sigma}(\tilde{u}) = \tilde{u}^{\sigma_n}, \quad u_i = N_{\Sigma_n|\Sigma}(\tilde{u}_i) = \tilde{u}_i^{\sigma_n}.$$

By (*), the elements $\tilde{u}^{\varphi^{-1}}$ and $\prod_i \tilde{u}_i^{\tau_i^{-1}}$ only differ by an element $\tilde{x} \in U_{\Sigma_n}$ such that $N_{\Sigma_n|\Sigma}(\tilde{x}) = 1$. Hence — again by (5.1) — they differ by an element of the form $\tilde{y}^{\sigma^{-1}}$, with $\tilde{y} \in U_{\Sigma_n}$. We may thus write

$$\tilde{u}^{\varphi^{-1}} = \tilde{y}^{\varphi_n^{-1}} \prod_i \tilde{u}_i^{\tau_i^{-1}} = (\tilde{y}^{\varphi_n})^{\varphi^{-1}} \prod_i \tilde{u}_i^{\tau_i^{-1}}.$$

Applying N gives $N(\tilde{u})^{\varphi^{-1}} = N(\tilde{y}^{\varphi_n})^{\varphi^{-1}}$, so that

$$N(\tilde{u}) = N(\tilde{y}^{\varphi_n}) \cdot z,$$

for some $z \in U_{\tilde{K}}$ such that $z^{\varphi^{-1}} = 1$; therefore $z^{\varphi} = z$, and $z \in U_K$. Finally, applying σ_n and putting $y = \tilde{y}^{\sigma_n} = N_{\Sigma_n|\Sigma}(\tilde{y}) \in U_{\Sigma}$, we obtain, observing $n = [M : K]$ and using (5.3), that

$$\begin{aligned} N(u) &= N(\tilde{u})^{\sigma_n} = N(\tilde{y}^{\varphi_n})^{\sigma_n} z^{\sigma_n} = N(y^{\varphi_n}) z^n \\ &= N_{\Sigma|K}(y) N_{M|K}(z) \in N_{M|K} U_M. \end{aligned}$$

□

(5.5) Proposition. *The reciprocity map*

$$r_{\tilde{L}|K} : \text{Frob}(\tilde{L}|K) \longrightarrow A_K / N_{\tilde{L}|K} A_{\tilde{L}}$$

is multiplicative.

Proof: Let $\sigma_1 \sigma_2 = \sigma_3$ be an equation in $\text{Frob}(\tilde{L}|K)$, $n_i = d_K(\sigma_i)$, Σ_i the fixed field of σ_i and $\pi_i \in A_{\Sigma_i}$ a prime element, for $i = 1, 2, 3$. We have to show that

$$N_{\Sigma_1|K}(\pi_1) N_{\Sigma_2|K}(\pi_2) \equiv N_{\Sigma_3|K}(\pi_3) \pmod{N_{\tilde{L}|K} A_{\tilde{L}}}.$$

Choose a fixed $\varphi \in G(\tilde{L}|K)$ such that $d_K(\varphi) = 1$ and put

$$\tau_i = \sigma_i^{-1} \varphi^{n_i} \in G(\tilde{L}|\tilde{K}).$$

From $\sigma_1 \sigma_2 = \sigma_3$ and $n_1 + n_2 = n_3$, we then deduce that

$$\tau_3 = \sigma_2^{-1} \sigma_1^{-1} \varphi^{n_2+n_1} = \sigma_2^{-1} \varphi^{n_2} (\varphi^{-n_2} \sigma_1 \varphi^{n_2})^{-1} \varphi^{n_1}.$$

Putting $\sigma_4 = \varphi^{-n_2} \sigma_1 \varphi^{n_2}$, $n_4 = d_K(\sigma_4) = n_1$, $\Sigma_4 = \Sigma_1^{\varphi^{n_2}}$, $\pi_4 = \pi_1^{\varphi^{n_2}} \in A_{\Sigma_4}$ and $\tau_4 = \sigma_4^{-1} \varphi^{n_4}$, we find $\tau_3 = \tau_2 \tau_4$ and

$$N_{\Sigma_4|K}(\pi_4) = N_{\Sigma_1|K}(\pi_1).$$

We may therefore pass to the congruence

$$N_{\Sigma_3|K}(\pi_3) \equiv N_{\Sigma_2|K}(\pi_2) N_{\Sigma_4|K}(\pi_4) \pmod{N_{\tilde{L}|K} A_{\tilde{L}}}$$

the proof of which uses the identity $\tau_3 = \tau_2\tau_4$. From (5.3), we have $N_{\Sigma_i|K}(\pi_i) = N(\pi_i^{\varphi_{n_i}})$. Thus, if we put

$$u = \pi_3^{\varphi_{n_3}} \pi_4^{-\varphi_{n_4}} \pi_2^{-\varphi_{n_2}},$$

then the congruence amounts simply to the relation $N(u) \in N_{\tilde{L}|K} A_{\tilde{L}}$. For this, however, lemma (5.4) gives us all that we need.

Since $\varphi_{n_i} \circ (\varphi - 1) = \varphi^{n_i} - 1$ and $\pi_i^{\varphi_{n_i}-1} = \pi_i^{\sigma_i^{-1}\varphi_{n_i}-1} = \pi_i^{\tau_i-1}$, we have

$$u^{\varphi-1} = \pi_3^{\tau_3-1} \pi_4^{1-\tau_4} \pi_2^{1-\tau_2}.$$

From the identity $\tau_3 = \tau_2\tau_4$, it follows that $(\tau_3 - 1) + (1 - \tau_2) + (1 - \tau_4) = (1 - \tau_2)(1 - \tau_4)$. Putting now

$$\pi_3 = u_3\pi_4, \quad \pi_2 = u_2^{-1}\pi_4, \quad \pi_4^{\tau_2} = u_4\pi_4, \quad u_i \in U_{\tilde{L}},$$

we obtain

$$u^{\varphi-1} = \prod_{i=2}^4 u_i^{\tau_i-1}.$$

For the element $x = u \bmod I_{G(\tilde{L}|\tilde{K})} U_{\tilde{L}} \in H_0(G(\tilde{L}|\tilde{K}), U_{\tilde{L}})$, this means that $x^{\varphi-1} = 1$, and so $x^{\varphi} = x$; then by (5.4), we do get $N(u) = N(x) \in N_{\tilde{L}|K} A_{\tilde{L}}$. $\square$

From the surjectivity of the mapping

$$\text{Frob}(\tilde{L}|K) \longrightarrow G(L|K)$$

and the fact that $N_{\tilde{L}|K} A_{\tilde{L}} \subseteq N_{L|K} A_L$, we now have the

(5.6) Proposition. *For every finite Galois extension $L|K$, there is a canonical homomorphism*

$$r_{L|K} : G(L|K) \longrightarrow A_K / N_{L|K} A_L$$

given by

$$r_{L|K}(\sigma) = N_{\Sigma|K}(\pi_{\Sigma}) \bmod N_{L|K} A_L,$$

where Σ is the fixed field of a preimage $\tilde{\sigma} \in \text{Frob}(\tilde{L}|K)$ of $\sigma \in G(L|K)$ and $\pi_{\Sigma} \in A_{\Sigma}$ is a prime element. It is called the **reciprocity homomorphism** of $L|K$.

Proof: We first show that the definition of $r_{L|K}(\sigma)$ is independent of the choice of the preimage $\tilde{\sigma} \in \text{Frob}(\tilde{L}|K)$ of σ . For this, let $\tilde{\sigma}' \in \text{Frob}(\tilde{L}|K)$ be another preimage, Σ' its fixed field and $\pi_{\Sigma'} \in A_{\Sigma'}$ a prime element. If $d_K(\tilde{\sigma}) = d_K(\tilde{\sigma}')$, then $\tilde{\sigma}|_{\tilde{K}} = \tilde{\sigma}'|_{\tilde{K}}$ and $\tilde{\sigma}|_L = \tilde{\sigma}'|_L$, so that $\tilde{\sigma} = \tilde{\sigma}'$, and there is nothing to show. However, if we have, say, $d_K(\tilde{\sigma}) < d_K(\tilde{\sigma}')$, then $\tilde{\sigma}' = \tilde{\sigma}\tilde{\tau}$ for some $\tilde{\tau} \in \text{Frob}(\tilde{L}|K)$, and $\tilde{\tau}|_L = 1$. The fixed field Σ'' of $\tilde{\tau}$ contains L , so $r_{\tilde{L}|K}(\tilde{\tau}) \equiv N_{\Sigma''|K}(\pi_{\Sigma''}) \equiv 1 \pmod{N_{L|K}A_L}$. It follows therefore that $r_{\tilde{L}|K}(\tilde{\sigma}') = r_{\tilde{L}|K}(\tilde{\sigma})r_{\tilde{L}|K}(\tilde{\tau}) = r_{\tilde{L}|K}(\tilde{\sigma})$.

The fact that the mapping is a homomorphism now follows directly from (5.5): if $\tilde{\sigma}_1, \tilde{\sigma}_2 \in \text{Frob}(\tilde{L}|K)$ are preimages of $\sigma_1, \sigma_2 \in G(L|K)$, then $\tilde{\sigma}_3 = \tilde{\sigma}_1\tilde{\sigma}_2$ is a preimage of $\sigma_3 = \sigma_1\sigma_2$. $\square$

The definition of the reciprocity map expresses the fundamental principle of class field theory to the effect that Frobenius automorphisms correspond to prime elements: the element $\sigma = \varphi_{\Sigma} \in G(\tilde{L}|\Sigma)$ is mapped to $\pi_{\Sigma} \in A_{\Sigma}$; for reasons of functoriality, the inclusion $G(\tilde{L}|\Sigma) \hookrightarrow G(\tilde{L}|K)$ corresponds to the norm map $N_{\Sigma|K} : A_{\Sigma} \rightarrow A_K$. So the definition of $r_{L|K}(\sigma)$ is already forced upon us by these requirements. This principle appears at its purest in the

(5.7) Proposition. *If $L|K$ is an unramified extension, then the reciprocity map*

$$r_{L|K} : G(L|K) \longrightarrow A_K/N_{L|K}A_L$$

is given by

$$r_{L|K}(\varphi_{L|K}) = \pi_K \pmod{N_{L|K}A_L},$$

and is an isomorphism.

Proof: In this case one has $\tilde{L} = \tilde{K}$ and $\varphi_K \in G(\tilde{K}|K)$ is a preimage of $\varphi_{L|K}$ with fixed field K , i.e., $r_{L|K}(\varphi_{L|K}) = \pi_K \pmod{N_{L|K}A_L}$. The fact that we have an isomorphism is seen from the composite

$$G(L|K) \longrightarrow A_K/N_{L|K}A_L \longrightarrow \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z},$$

with $n = [L : K]$, where the second map is induced by the valuation $v_K : A_K \rightarrow \mathbb{Z}$ because $v_K(N_{L|K}A_L) \subseteq n\mathbb{Z}$. It is an isomorphism, for if $v_K(a) \equiv 0 \pmod{n\mathbb{Z}}$, then $a = u\pi_K^{dn}$, and since $u = N_{L|K}(\varepsilon)$ for some $\varepsilon \in U_L$, by (5.1), we find $a = N_{L|K}(\varepsilon\pi_K^d) \equiv 1 \pmod{N_{L|K}A_L}$. On the side of the homomorphisms, the generators $\varphi_{L|K}$, $\pi_K \pmod{N_{L|K}A_L}$, and $1 \pmod{n\mathbb{Z}}$ correspond to each other, and everything is proved. $\square$

The reciprocity homomorphism $r_{L|K}$ exhibits the following functorial behaviour.

(5.8) Proposition. *Let $L|K$ and $L'|K'$ be finite Galois extensions, so that $K \subseteq K'$ and $L \subseteq L'$, and let $\sigma \in G$. Then we have the commutative diagrams*

$$\begin{array}{ccc}
 G(L'|K') & \xrightarrow{r_{L'|K'}} & A_{K'}/N_{L'|K'}A_{L'} \\
 \downarrow & & \downarrow N_{K'|K} \\
 G(L|K) & \xrightarrow{r_{L|K}} & A_K/N_{L|K}A_L
 \end{array}
 \qquad
 \begin{array}{ccc}
 G(L|K) & \xrightarrow{r_{L|K}} & A_K/N_{L|K}A_L \\
 \downarrow \sigma^* & & \downarrow \sigma \\
 G(L^\sigma|K^\sigma) & \xrightarrow{r_{L^\sigma|K^\sigma}} & A_{K^\sigma}/N_{L^\sigma|K^\sigma}A_{L^\sigma}
 \end{array}$$

where the vertical arrows on the left are given by $\sigma' \mapsto \sigma'|_L$, resp. by the conjugation $\tau \mapsto \sigma^{-1}\tau\sigma$.

Proof: Let $\sigma' \in G(L'|K')$ and $\sigma = \sigma'|_L \in G(L|K)$. If $\tilde{\sigma}' \in \text{Frob}(\tilde{L}'|K')$ is a preimage of σ' , then $\tilde{\sigma} = \tilde{\sigma}'|_{\tilde{L}} \in \text{Frob}(\tilde{L}|K)$ is a preimage of σ such that $d_K(\tilde{\sigma}) = f_{K'|K}d_{K'}(\tilde{\sigma}') \in \mathbb{N}$. Let Σ' be the fixed field of $\tilde{\sigma}'$. Then $\Sigma = \Sigma' \cap \tilde{L} = \Sigma' \cap \tilde{\Sigma}$ is the fixed field of $\tilde{\sigma}$ and $f_{\Sigma'|\Sigma} = 1$. If now $\pi_{\Sigma'} \in A_{\Sigma'}$ is a prime element of Σ' , then $\pi_\Sigma = N_{\Sigma'|\Sigma}(\pi_{\Sigma'}) \in A_\Sigma$ is a prime element of Σ . The commutativity of the diagram on the left therefore follows from the equality of norms

$$N_{\Sigma|K}(\pi_\Sigma) = N_{\Sigma|K}(N_{\Sigma'|\Sigma}(\pi_{\Sigma'})) = N_{\Sigma'|K}(\pi_{\Sigma'}) = N_{K'|K}(N_{\Sigma'|K'}(\pi_{\Sigma'})).$$

On the other hand, let $\tau \in G(L|K)$, and let $\tilde{\tau}$ be a preimage in $\text{Frob}(\tilde{L}|K)$ with fixed field Σ , and $\hat{\tau} \in G$ a lifting of $\tilde{\tau}$ to $\bar{k}$. Then Σ^σ is the fixed field of $\sigma^{-1}\hat{\tau}\sigma|_{\tilde{L}^\sigma}$, and if $\pi \in A_\Sigma$ is a prime element of Σ , then $\pi^\sigma \in A_{\Sigma^\sigma}$ is a prime element of Σ^σ . The commutativity of the diagram on the right therefore follows from the equality of norms

$$N_{\Sigma^\sigma|K^\sigma}(\pi^\sigma) = N_{\Sigma|K}(\pi)^\sigma. \quad \square$$

Another very interesting functorial property of the reciprocity map is obtained via the *transfer* (*Verlagerung* in German). For an arbitrary group G , let G' denote the commutator subgroup and write

$$G^{ab} = G/G'$$

for the maximal abelian quotient group. If then $H \subseteq G$ is a subgroup of finite index, we have a canonical homomorphism

$$\text{Ver} : G^{ab} \longrightarrow H^{ab},$$

which is called **transfer from G to H** . This homomorphism is defined as follows (see [75], chap. IV, § 1).

Let R be a system of representatives for the left cosets of H in G , $G = RH$, $1 \in R$. If $\sigma \in G$ we write, for every $\rho \in R$,

$$\sigma\rho = \rho'\sigma_\rho, \quad \sigma_\rho \in H, \quad \rho' \in R,$$

and we define

$$\text{Ver}(\sigma \bmod G') = \prod_{\rho \in R} \sigma_\rho \bmod H'.$$

Another description of the transfer results from the double coset decomposition

$$G = \bigcup_{\tau} (\sigma)\tau H$$

of G in terms of the subgroups (σ) and H . Letting $f(\tau)$ denote the smallest natural number such that $\sigma_\tau = \tau^{-1}\sigma^{f(\tau)}\tau \in H$, one has $H \cap (\tau^{-1}\sigma\tau) = (\sigma_\tau)$, and we find that

$$\text{Ver}(\sigma \bmod G') = \prod_{\tau} \sigma_\tau \bmod H'.$$

This formula is obtained from the one above by choosing for R the set $\{\sigma^i\tau \mid i = 1, \dots, f(\tau)\}$. Applying this to the reciprocity homomorphism

$$r_{L|K} : G(L|K)^{ab} \longrightarrow A_K/N_{L|K}A_L$$

we get the

(5.9) Proposition. *Let $L|K$ be a finite Galois extension and K' an intermediate field. Then we have the commutative diagram*

$$\begin{array}{ccc} G(L|K')^{ab} & \xrightarrow{r_{L|K'}} & A_{K'}/N_{L|K'}A_L \\ \uparrow \text{Ver} & & \uparrow \\ G(L|K)^{ab} & \xrightarrow{r_{L|K}} & A_K/N_{L|K}A_L, \end{array}$$

where the arrow on the right is induced by inclusion.

Proof: Let us write temporarily $G = G(\tilde{L}|K)$ and $H = G(\tilde{L}|K')$. Let $\sigma \in G(L|K)$, and let $\tilde{\sigma}$ be a preimage in $\text{Frob}(\tilde{L}|K)$ with fixed field Σ and $S = G(\tilde{L}|\Sigma) = (\tilde{\sigma})$. We consider the double coset decomposition $G = \bigcup_{\tau} S\tau H$ and put $S_\tau = \tau^{-1}S\tau \cap H$ and $\tilde{\sigma}_\tau = \tau^{-1}\tilde{\sigma}^{f(\tau)}\tau$ as above. Let $\bar{G} = G(L|K)$, $\bar{H} = G(L|K')$, $\bar{S} = (\sigma)$, $\bar{\tau} = \tau|_L$ and $\sigma_\tau = \tilde{\sigma}_\tau|_L$. Then we obviously have

$$\bar{G} = \bigcup_{\tau} \bar{S} \bar{\tau} \bar{H},$$

and therefore

$$\text{Ver}(\sigma \bmod G(L|K)') = \prod_{\tau} \sigma_{\tau} \bmod G(L|K)'. \quad \square$$

For every τ , let ω_{τ} vary over a system of right coset representatives of H/S_{τ} . Then one has

$$H = \bigcup_{\omega_{\tau}} S_{\tau} \omega_{\tau} \quad \text{and} \quad G = \bigcup_{\tau, \omega_{\tau}} S_{\tau} \omega_{\tau}.$$

Let Σ_{τ} be the fixed field of $\tilde{\sigma}_{\tau}$, i.e., the fixed field of S_{τ} . Σ^{τ} is the fixed field of $\tau^{-1}\tilde{\sigma}\tau$ so that $\Sigma_{\tau}|\Sigma^{\tau}$ is the unramified subextension of degree $f(\tau)$ in $\tilde{L}|\Sigma^{\tau}$. If now $\pi \in A_{\Sigma}$ is a prime element of Σ , then $\pi^{\tau} \in A_{\Sigma^{\tau}}$ is a prime element of Σ^{τ} , and thus also of Σ_{τ} . In view of the above double coset decomposition, we therefore find

$$N_{\Sigma|K}(\pi) = \prod_{\tau, \omega_{\tau}} \pi^{\tau \omega_{\tau}} = \prod_{\tau} \left(\prod_{\omega_{\tau}} (\pi^{\tau})^{\omega_{\tau}} \right) = \prod_{\tau} N_{\Sigma_{\tau}|K'}(\pi^{\tau}),$$

and since $\tilde{\sigma}_{\tau} \in \text{Frob}(\tilde{L}|K')$ is a preimage of $\sigma_{\tau} \in G(L|K')$, it follows that

$$r_{L|K}(\sigma) \equiv \prod_{\tau} r_{L|K'}(\sigma_{\tau}) \equiv r_{L|K'}\left(\prod_{\tau} \sigma_{\tau}\right) \equiv r_{L|K'}\left(\text{Ver}(\sigma \bmod G(L|K)')\right).$$

□

Exercise 1. Let $L|K$ be abelian and totally ramified, and let π be a prime element of A_L . If then $\sigma \in G(L|K)$ and

$$y^{\varphi-1} = \pi^{\sigma-1}$$

with $y \in U_L$, then $r_{L|K}(\sigma) = N(y) \bmod N_{L|K} A_L$, where $N = N_{\tilde{L}|\tilde{K}}$ (*B.DWORK*, see [122], chap. XIII, §5).

Exercise 2. Generalize the theory developed so far in the following way. Let P be a set of prime numbers and let G be a pro- P -group, i.e., a profinite group all of whose quotients G/N by open normal subgroups N have order divisible only by primes in P .

Let $d : G \rightarrow \mathbb{Z}_P$ be a surjective homomorphism onto the group $\mathbb{Z}_P = \prod_{p \in P} \mathbb{Z}_p$, and let A be a G -module. A **henselian P -valuation** with respect to d is by definition a homomorphism

$$v : A_k \rightarrow \mathbb{Z}_P$$

which satisfies the following properties:

- (i) $v(A_K) = Z \supseteq \mathbb{Z}$ and $Z/nZ \cong \mathbb{Z}/n\mathbb{Z}$ for all natural numbers n which are divisible only by primes in P ,
- (ii) $v(N_{K|k} A_K) = f_K Z$ for all finite extensions $K|k$, where $f_K = (d(G) : d(G_K))$.

Under the hypothesis that $H^i(G(L|K), U_L) = 1$ for $i = 0, -1$, for all unramified extensions $L|K$, prove the existence of a canonical reciprocity homomorphism $r_{L|K} : G(L|K)^{ab} \rightarrow A_K/N_{L|K} A_L$ for every finite Galois extension $L|K$.

Exercise 3. Let $d : G \rightarrow \widehat{\mathbb{Z}}$ be a surjective homomorphism, A a G -module which satisfies axiom (5.1), and let $v : A_k \rightarrow \widehat{\mathbb{Z}}$ be a henselian valuation with respect to d .

Let $K|k$ be a finite extension and let $\text{spec}(K)$ be the set of **microprimes** of K (see § 4, exercise 1–5). Define a canonical mapping

$$r_K : \text{spec}(K) \rightarrow A_K/N_{\bar{k}|K}^- A_{\bar{k}}^-,$$

and show that, for a finite extension, the diagram

$$\begin{array}{ccc} \text{spec}(L) & \xrightarrow{r_L} & A_L/N_{\bar{k}|L}^- A_{\bar{k}}^- \\ \pi \downarrow & & \downarrow N_{L|K} \\ \text{spec}(K) & \xrightarrow{r_K} & A_K/N_{\bar{k}|K}^- A_{\bar{k}}^- \end{array}$$

commutes. Show furthermore that, for every finite Galois extension $L|K$, r_K induces the reciprocity isomorphism

$$r_{L|K} : G(L|K) \rightarrow A_K/N_{L|K} A_L.$$

Hint: Let $\varphi \in G_K$ be an element such that $d_K(\varphi) \in \mathbb{N}$. Let Σ be the fixed field of φ and

$$\widehat{A}_\Sigma = \varprojlim_{\alpha} A_{K_\alpha},$$

where $K_\alpha|K$ varies over the finite subextensions of $\Sigma|K$, and where the projective limit is taken with respect to the norm maps $N_{K_\beta|K_\alpha} : A_{K_\beta} \rightarrow A_{K_\alpha}$. Then there is a surjective homomorphism $v_\Sigma : \widehat{A}_\Sigma \rightarrow Z$ and a homomorphism $N_{\Sigma|K} : \widehat{A}_\Sigma \rightarrow A_K$.

§ 6. The General Reciprocity Law

We now impose on the continuous G -module A the following condition.

(6.1) Class Field Axiom. For every cyclic extension $L|K$ one has

$$\#H^i(G(L|K), A_L) = \begin{cases} [L : K] & \text{for } i = 0, \\ 1 & \text{for } i = -1. \end{cases}$$

Among the cyclic extensions there are in particular the unramified ones. For them the above condition amounts precisely to requiring axiom (5.1), so that one has

(6.2) Proposition. For a finite unramified extension $L|K$, one has

$$H^i(G(L|K), U_L) = 1 \quad \text{for } i = 0, -1.$$

Proof: Since $L|K$ is unramified, a prime element π_K of A_K is also a prime element of A_L . As $H^{-1}(G(L|K), A_L) = 1$, every element $u \in U_L$ such that $N_{L|K}(u) = 1$ is of the form $u = a^{\sigma^{-1}}$, with $a \in A_L$, $\sigma = \varphi_{L|K}$. So writing $a = \varepsilon \pi_K^m$, $\varepsilon \in U_L$, we obtain $u = \varepsilon^{\sigma^{-1}}$. This shows that $H^{-1}(G(L|K), U_L) = 1$.

On the other hand, the homomorphism $v_K : A_K \rightarrow Z$ gives rise to a homomorphism

$$v_K : A_K / N_{L|K} A_L \longrightarrow Z/nZ = \mathbb{Z}/n\mathbb{Z},$$

where $n = [L : K] = f_{L|K}$, because $v_K(N_{L|K} A_L) = f_{L|K} Z = nZ$. This homomorphism is surjective as $v_K(\pi_K \bmod N_{L|K} A_L) = 1 \bmod nZ$, and it is bijective as $\#A_K / N_{L|K} A_L = n$. If now $u \in U_K$, then we have $u = N_{L|K}(a)$, with $a \in A_L$, since $v_K(u) = 0$. But $0 = v_K(u) = v_K(N_{L|K}(a)) = nv_L(a)$, so we get in fact $a \in U_L$. This proves that $H^0(G(L|K), U_L) = 1$. $\square$

By definition, a **class field theory** is a pair of homomorphisms

$$(d : G \rightarrow \widehat{\mathbb{Z}}, v : A \rightarrow \widehat{\mathbb{Z}}),$$

where A is a G -module which satisfies axiom (6.1), d is a surjective continuous homomorphism, and v is a henselian valuation. From proposition (6.2) and §5, we obtain for every finite Galois extension $L|K$, the reciprocity homomorphism

$$r_{L|K} : G(L|K)^{ab} \longrightarrow A_K / N_{L|K} A_L.$$

But the class field axiom yields moreover the following theorem, which represents the main theorem of class field theory, and which we will call the **general reciprocity law**.

(6.3) Theorem. *For every finite Galois extension $L|K$, the homomorphism*

$$r_{L|K} : G(L|K)^{ab} \longrightarrow A_K / N_{L|K} A_L$$

is an isomorphism.

Proof: If $M|K$ is a Galois subextension of $L|K$, we get from (5.8) the commutative exact diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & G(L|M) & \longrightarrow & G(L|K) & \longrightarrow & G(M|K) \longrightarrow 1 \\ & & \downarrow r_{L|M} & & \downarrow r_{L|K} & & \downarrow r_{M|K} \\ & & A_M / N_{L|M} A_L & \xrightarrow{N_{M|K}} & A_K / N_{L|K} A_L & \longrightarrow & A_K / N_{M|K} A_M \longrightarrow 1. \end{array}$$

We use this diagram to make three reductions.

First reduction. We may assume that $G(L|K)$ is abelian. For if the theorem is proved in this case, then, putting $M = L^{ab}$ the maximal abelian subextension of $L|K$, we find $G(L|K)^{ab} = G(M|K)$, and the commutator subgroup $G(L|M)$ of $G(L|K)$ is precisely the kernel of $r_{L|K}$, i.e., $G(L|K)^{ab} \rightarrow A_K/N_{L|K}A_L$ is injective. The surjectivity follows by induction on the degree. Indeed, in the case where $G(L|K)$ is solvable, one has either $M = L$ or $[L : M] < [L : K]$, and if $r_{M|K}$ and $r_{L|M}$ are surjective, then so is $r_{L|K}$. In the general case, let M be the fixed field of a p -Sylow subgroup. $M|K$ need not be Galois, but we may use the left part of the diagram, where $r_{L|M}$ is surjective. It then suffices to show that the image of $N_{M|K}$ is the p -Sylow subgroup S_p of $A_K/N_{L|K}A_L$. That this holds true for all p amounts to saying that $r_{L|K}$ is surjective. Now the inclusion $A_K \subseteq A_M$ induces a homomorphism

$$i : A_K/N_{L|K}A_L \longrightarrow A_M/N_{L|M}A_L,$$

such that $N_{M|K} \circ i = [M : K]$. As $([M : K], p) = 1$, $S_p \xrightarrow{[M:K]} S_p$ is surjective, so S_p lies in the image of $N_{M|K}$, and therefore of $r_{L|K}$.

Second reduction. We may assume that $L|K$ is cyclic. For if $M|K$ varies over all cyclic subextensions of $L|K$, then the diagram shows that the kernel of $r_{L|K}$ lies in the kernel of the map $G(L|K) \rightarrow \prod_M G(M|K)$. Since $G(L|K)$ is abelian, this map is injective and hence the same is true of $r_{L|K}$. Choosing a proper cyclic subextension $M|K$ of $L|K$, surjectivity follows by induction on the degree as in the first reduction for solvable extensions.

Third reduction. Let $L|K$ be cyclic. We may assume that $f_{L|K} = 1$. To see this, let $M = L \cap \tilde{K}$ be the maximal unramified subextension of $L|K$. Then $f_{L|M} = 1$ and $r_{M|K}$ is an isomorphism by (5.7). In the bottom sequence of our diagram, the map $N_{M|K}$ is injective because the groups in this sequence have the respective orders $[L : M]$, $[L : K]$, $[M : K]$ by axiom (6.1). Therefore $r_{L|K}$ is an isomorphism if $r_{L|M}$ is.

Now let $L|K$ be cyclic and totally ramified, i.e., $f_{L|K} = 1$. Let σ be a generator of $G(L|K)$. We view σ via the isomorphism $G(L|K) \cong G(\tilde{L}|\tilde{K})$ as an element of $G(\tilde{L}|\tilde{K})$, and obtain the element $\tilde{\sigma} = \sigma\varphi_L \in \text{Frob}(\tilde{L}|\tilde{K})$, which is a preimage of $\sigma \in G(L|K)$ such that $d_K(\tilde{\sigma}) = d_K(\varphi_L) = f_{L|K} = 1$. We thus find for the fixed field $\Sigma|K$ of $\tilde{\sigma}$ that $f_{\Sigma|K} = 1$, and so $\Sigma \cap \tilde{K} = K$. Let $M|K$ be a finite Galois subextension of $\tilde{L}|K$ containing Σ and L , let $M^0 = M \cap \tilde{K}$ be the maximal unramified subextension of $M|K$, and put $N = N_{M|M^0}$. As $f_{\Sigma|K} = f_{L|K} = 1$, one finds $N|_{A_\Sigma} = N_{\Sigma|K}$, $N|_{A_L} = N_{L|K}$ (see the proof of (5.3)).

For the injectivity of $r_{L|K}$, we have to prove this: if $r_{L|K}(\sigma^k) = 1$, where $0 \leq k < n = [L : K]$, then $k = 0$.

In order to do this, let $\pi_\Sigma \in A_\Sigma$, $\pi_L \in A_L$ be prime elements. Since $\Sigma, L \subseteq M$, π_Σ and π_L are both prime elements of M . Putting $\pi_\Sigma^k = u\pi_L^k$, $u \in U_M$, we obtain

$$r_{L|K}(\sigma^k) \equiv N(\pi_\Sigma^k) \equiv N(u) \cdot N(\pi_L^k) \equiv N(u) \pmod{N_{L|K} A_L}.$$

From $r_{L|K}(\sigma^k) = 1$, it thus follows that $N(u) = N(v)$ for some $v \in U_L$, so that $N(u^{-1}v) = 1$. From axiom (6.1), we may write $u^{-1}v = a^{\sigma^{-1}}$ for some $a \in A_M$, and find in A_M the equation

$$(\pi_L^k v)^{\sigma^{-1}} = (\pi_L^k v)^{\tilde{\sigma}^{-1}} = (\pi_\Sigma^k u^{-1} v)^{\tilde{\sigma}^{-1}} = (a^{\sigma^{-1}})^{\tilde{\sigma}^{-1}} = (a^{\tilde{\sigma}^{-1}})^{\sigma^{-1}},$$

and so $x = \pi_L^k v a^{1-\tilde{\sigma}} \in A_{M^0}$. Now $v_{M^0}(x) \in \widehat{\mathbb{Z}}$ and $nv_{M^0}(x) = v_M(x) = k$ imply that one has $k = 0$, and so $r_{L|K}$ is injective. The surjectivity then follows from (6.1). $\square$

The inverse of the mapping $r_{L|K} : G(L|K)^{ab} \rightarrow A_K/N_{L|K} A_L$ gives, for every finite Galois extension $L|K$, a surjective homomorphism

$$(\cdot, L|K) : A_K \longrightarrow G(L|K)^{ab}$$

with kernel $N_{L|K} A_L$. This map is called the **norm residue symbol** of $L|K$. From (5.8) and (5.9) we have the

(6.4) Proposition. *Let $L|K$ and $L'|K'$ be finite Galois extensions such that $K \subseteq K'$ and $L \subseteq L'$, and let $\sigma \in G$. Then we have the commutative diagrams*

$$\begin{array}{ccc} A_{K'} & \xrightarrow{(\cdot, L'|K')} & G(L'|K')^{ab} \\ \downarrow N_{K'|K} & & \downarrow \\ A_K & \xrightarrow{(\cdot, L|K)} & G(L|K)^{ab}, \end{array} \quad \begin{array}{ccc} A_K & \xrightarrow{(\cdot, L|K)} & G(L|K)^{ab} \\ \downarrow \sigma & & \downarrow \sigma^* \\ A_{K\sigma} & \xrightarrow{(\cdot, L^\sigma|K^\sigma)} & G(L^\sigma|K^\sigma)^{ab}, \end{array}$$

and if $K' \subseteq L$, we have the commutative diagram

$$\begin{array}{ccc} A_{K'} & \xrightarrow{(\cdot, L|K')} & G(L|K')^{ab} \\ \uparrow & & \uparrow \text{Ver} \\ A_K & \xrightarrow{(\cdot, L|K)} & G(L|K)^{ab}. \end{array}$$

The definition of the norm residue symbol automatically extends to infinite Galois extensions $L|K$. For if $L_i|K$ varies over the finite Galois subextensions, then

$$G(L|K)^{ab} = \varprojlim_i G(L_i|K)^{ab}$$

(see § 2, exercise 6). As $(a, L_{i'}|K)|_{L_i^{ab}} = (a, L_i|K)$ for $L_{i'} \supseteq L_i$, the individual norm residue symbols $(a, L_i|K)$, $a \in A_K$, determine an element

$$(a, L|K) \in G(L|K)^{ab}.$$

In the special case of the extension $\tilde{K}|K$, we find the following intimate connection between the maps d_K , v_K , and $(\ , \tilde{K}|K)$.

(6.5) Proposition. *One has*

$$(a, \tilde{K}|K) = \varphi_K^{v_K(a)}, \quad \text{and thus} \quad d_K \circ (\ , \tilde{K}|K) = v_K.$$

Proof: Let $L|K$ be the subextension of $\tilde{K}|K$ of degree f . As $Z/fZ = \mathbb{Z}/f\mathbb{Z}$, we have $v_K(a) = n + fz$, with $n \in \mathbb{Z}$, $z \in \mathbb{Z}$; that is, $a = u\pi_K^n b^f$, with $u \in U_K$, $b \in A_K$. From (5.7), we obtain

$$(a, \tilde{K}|K)|_L = (a, L|K) = (\pi_K, L|K)^n (b, L|K)^f = \varphi_{L|K}^n = \varphi_K^{v_K(a)}|_L.$$

Thus we must have $(a, \tilde{K}|K) = \varphi_K^{v_K(a)}$. □

The main goal of field theory is to classify all algebraic extensions of a given field K . The law governing the constitution of extensions of K is hidden in the inner structure of the base field K itself, and should therefore be expressed in terms of entities directly associated with it. Class field theory solves this problem as far as the abelian extensions of K are concerned. It establishes a 1–1-correspondence between these extensions and certain subgroups of A_K . More precisely, this is done as follows.

For every field K , we equip the group A_K with a topology by declaring the cosets $aN_{L|K}A_L$ to be a basis of neighbourhoods of $a \in A_K$, where $L|K$ varies over all finite Galois extensions of K . We call this topology the **norm topology** of A_K .

(6.6) Proposition. (i) *The open subgroups of A_K are precisely the closed subgroups of finite index.*

(ii) *The valuation $v_K : A_K \rightarrow \widehat{\mathbb{Z}}$ is continuous.*

(iii) If $L|K$ is a finite extension, then $N_{L|K} : A_L \rightarrow A_K$ is continuous.

(iv) A_K is Hausdorff if and only if the group

$$A_K^0 = \bigcap_L N_{L|K} A_L$$

of universal norms is trivial.

Proof: (i) If $\mathcal{N}$ is a subgroup of A_K , then

$$\mathcal{N} = A_K \setminus \bigcup_{a\mathcal{N} \neq \mathcal{N}} a\mathcal{N}.$$

Now, if $\mathcal{N}$ is open, so are all cosets $a\mathcal{N}$, so that $\mathcal{N}$ is closed, and since $\mathcal{N}$ has to contain one of the neighbourhoods $N_{L|K} A_L$ of the basis of neighbourhoods of 1, $\mathcal{N}$ is also of finite index. If, conversely, $\mathcal{N}$ is closed and of finite index, then the union of the finitely many cosets $a\mathcal{N} \neq \mathcal{N}$ is closed, and so $\mathcal{N}$ is open.

(ii) The groups $f\widehat{\mathbb{Z}}$, $f \in \mathbb{N}$, form a basis of neighbourhoods of $0 \in \widehat{\mathbb{Z}}$ (see §2), and if $L|K$ is the unramified extension of degree f , then it follows from (4.7) that

$$v_K(N_{L|K} A_L) = f v_L(A_L) \subseteq f\widehat{\mathbb{Z}},$$

which shows the continuity of v_K .

(iii) Let $N_{M|K} A_M$ be an open neighbourhood of $1 \in A_K$. Then

$$N_{L|K}(N_{ML|L} A_{ML}) = N_{ML|K} A_{ML} \subseteq N_{M|K} A_M,$$

which shows the continuity of $N_{L|K}$.

(iv) is self-evident. □

The finite abelian extensions $L|K$ are now classified as follows.

(6.7) Theorem. Associating

$$L \mapsto \mathcal{N}_L = N_{L|K} A_L$$

sets up a 1–1-correspondence between the finite abelian extensions $L|K$ and the open subgroups $\mathcal{N}$ of A_K . Furthermore, one has

$$L_1 \subseteq L_2 \iff \mathcal{N}_{L_1} \supseteq \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}, \quad \mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1} \mathcal{N}_{L_2}.$$

The field L corresponding to the subgroup $\mathcal{N}$ of A_K is called the **class field** associated with $\mathcal{N}$. By (6.3), it satisfies

$$G(L|K) \cong A_K / \mathcal{N}.$$

Proof of (6.7): If L_1 and L_2 are abelian extensions of K , then the transitivity of the norm implies $\mathcal{N}_{L_1 L_2} \subseteq \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}$. If, conversely, $a \in \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}$, then the element $(a, L_1 L_2 | K) \in G(L_1 L_2 | K)$ projects trivially onto $G(L_i | K)$, that is, $(a, L_i | K) = 1$ for $i = 1, 2$. Thus $(a, L_1 L_2 | K) = 1$, i.e., $a \in \mathcal{N}_{L_1 L_2}$. We therefore have $\mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2}$, and so

$$\begin{aligned} \mathcal{N}_{L_1} \supseteq \mathcal{N}_{L_2} &\iff \mathcal{N}_{L_1} \cap \mathcal{N}_{L_2} = \mathcal{N}_{L_1 L_2} = \mathcal{N}_{L_2} \iff [L_1 L_2 : K] \\ &= [L_2 : K] \iff L_1 \subseteq L_2. \end{aligned}$$

This shows the injectivity of the correspondence $L \mapsto \mathcal{N}_L$.

If $\mathcal{N}$ is any open subgroup, then it contains the norm group $\mathcal{N}_L = N_{L|K} A_L$ of some Galois extension $L|K$. (6.3) implies that $\mathcal{N}_L = \mathcal{N}_{L^{ab}}$, so we may assume $L|K$ to be abelian. But $(\mathcal{N}, L|K) = G(L|L')$ for some intermediate field L' of $L|K$. Since $\mathcal{N} \supseteq \mathcal{N}_L$, the group $\mathcal{N}$ is the full preimage of $G(L|L')$ under the map $(\cdot, L|K) : A_K \rightarrow G(L|K)$, and thus it is the full kernel of $(\cdot, L'|K) : A_K \rightarrow G(L'|K)$. Thus $\mathcal{N} = \mathcal{N}_{L'}$. This shows that the correspondence $L \mapsto \mathcal{N}_L$ is surjective.

Finally, the equality $\mathcal{N}_{L_1 \cap L_2} = \mathcal{N}_{L_1} \mathcal{N}_{L_2}$ is obtained as follows. $L_1 \cap L_2 \subseteq L_i$ implies that $\mathcal{N}_{L_1 \cap L_2} \supseteq \mathcal{N}_{L_i}$, $i = 1, 2$, and thus

$$\mathcal{N}_{L_1 \cap L_2} \supseteq \mathcal{N}_{L_1} \mathcal{N}_{L_2}.$$

As $\mathcal{N}_{L_1} \mathcal{N}_{L_2}$ is open, we have just shown that $\mathcal{N}_{L_1} \mathcal{N}_{L_2} = \mathcal{N}_L$ for some finite abelian extension $L|K$. But $\mathcal{N}_{L_i} \subseteq \mathcal{N}_L$ implies $L \subseteq L_1 \cap L_2$, so that

$$\mathcal{N}_{L_1} \mathcal{N}_{L_2} = \mathcal{N}_L \supseteq \mathcal{N}_{L_1 \cap L_2}. \quad \square$$

Exercise 1. Let n be a natural number, and assume the group $\mu_n = \{\xi \in A \mid \xi^n = 1\}$ is cyclic of order n , and $A \subseteq A^n$. Let K be a field such that $\mu_n \subseteq A_K$, and let $L|K$ be the maximal abelian extension of exponent n . If $L|K$ is finite, then one has $N_{L|K} A_L = A_K^n$.

Exercise 2. Under the hypotheses of exercise 1, Kummer theory and class field theory yield, via Pontryagin duality $G(L|K) \times \text{Hom}(G(L|K), \mu_n) \rightarrow \mu_n$, a nondegenerate bilinear mapping (the abstract “Hilbert symbol”)

$$(\cdot, \cdot) : A_K / A_K^n \times A_K / A_K^n \longrightarrow \mu_n.$$

Exercise 3. Let p be a prime number and $(d : G \rightarrow \mathbb{Z}_p, v : A_k \rightarrow \mathbb{Z}_p)$ a p -class field theory in the sense of § 5, exercise 2. Let $d' : G \rightarrow \mathbb{Z}_p$ be another surjective homomorphism, and $\widehat{K}'|K$ the $\mathbb{Z}_p$ -extension defined by d' . Let $v' : A_k \rightarrow \mathbb{Z}_p$ be the composite of

$$A_k \xrightarrow{(\cdot, \widehat{K}'|K)} G(\widehat{K}'|K) \xrightarrow{d'} \mathbb{Z}_p.$$

Then (d', v') is also a p -class field theory. The norm residue symbols associated to (d, v) and (d', v') coincide. (No analogous statement holds in the case of $\widehat{\mathbb{Z}}$ -class field theories $(d : G \rightarrow \widehat{\mathbb{Z}}, v : A_k \rightarrow \widehat{\mathbb{Z}})$.)

Exercise 4. (Generalization to infinite extensions.) Let $(d : G \rightarrow \widehat{\mathbb{Z}}, v : A_k \rightarrow \widehat{\mathbb{Z}})$ be a class field theory. We assume that the kernel U_k of $v_k : A_k \rightarrow \widehat{\mathbb{Z}}$ is compact for every finite extension $K|k$. For an infinite extension $K|k$, put

$$\widehat{A}_K = \varprojlim A_{K_\alpha},$$

where $K_\alpha|k$ varies over the finite subextensions of $K|k$ and the projective limit is taken with respect to the norm maps $N_{K_\beta|K_\alpha} : A_{K_\beta} \rightarrow A_{K_\alpha}$. Show:

1) For every (finite or infinite) extension $L|K$, one has a norm map

$$N_{L|K} : \widehat{A}_L \rightarrow \widehat{A}_K,$$

and if $L|K$ is finite, there is an injection $i_{L|K} : \widehat{A}_K \rightarrow \widehat{A}_L$. If furthermore $L|K$ is Galois, then one has $\widehat{A}_K \cong \widehat{A}_L^{G(L|K)}$.

2) For every extension $K|k$ with finite inertia degree $f_K = [K \cap \bar{k} : k]$, (d, v) induces a class field theory $(d_K : G_K \rightarrow \widehat{\mathbb{Z}}, v_K : \widehat{A}_K \rightarrow \widehat{\mathbb{Z}})$.

3) If $K \subseteq K'$ are extensions of k with $f_K, f_{K'} < \infty$, and $L|K$ and $L'|K'$ are (finite or infinite) Galois extensions with $L \subseteq L'$, then one has a commutative diagram

$$\begin{array}{ccc} \widehat{A}_{K'} & \xrightarrow{(\cdot, L'|K')} & G(L'|K')^{ab} \\ \downarrow N_{K'|K} & & \downarrow \\ \widehat{A}_K & \xrightarrow{(\cdot, L|K)} & G(L|K)^{ab}. \end{array}$$

Exercise 5. If $L|K$ is a finite Galois extension, then G_L^{ab} is a $G(L|K)$ -module in a canonical way, and the transfer from G_K to G_L is a homomorphism

$$\text{Ver} : G_K^{ab} \rightarrow (G_L^{ab})^{G(L|K)}.$$

Exercise 6. (Tautological class field theory.) Assume that the profinite group G satisfies the condition: for every finite Galois extension,

$$\text{Ver} : G_K^{ab} \rightarrow (G_L^{ab})^{G(L|K)}$$

is an isomorphism. (These are the profinite groups of “strict cohomological dimension 2” (see [145], chap. III, th. (3.6.4)).) Put $A_K = G_K^{ab}$ and form the direct limit $A = \varinjlim A_K$ via the transfer. Then A_K is identified with A^{G_K} .

Show that for every cyclic extension $L|K$ one has

$$\#H^i(G(L|K), A_L) = \begin{cases} [L : K] & \text{for } i = 0, \\ 1 & \text{for } i = -1, \end{cases}$$

and that for every surjective homomorphism $d : G \rightarrow \widehat{\mathbb{Z}}$, the induced map $v : A_k = G^{ab} \rightarrow \widehat{\mathbb{Z}}$ is a henselian valuation with respect to d . The corresponding reciprocity map $r_{L|K} : G(L|K) \rightarrow A_K/N_{L|K}A_L$ is essentially the identity.

Abstract class field theory acquires a much broader range of applications if it is generalized as follows.

Exercise 7. Let G be a profinite group and $B(G)$ the category of finite G -sets, i.e., of finite sets X with a continuous G -operation. Show that the *connected*, i.e., transitive G -sets in $B(G)$ are, up to isomorphism, the sets G/G_K , where G_K is an open subgroup of G , and G operates via multiplication on the left.

If X is a finite G -set and $x \in X$, then

$$\pi_1(X, x) = G_x = \{\sigma \in G \mid \sigma x = x\}$$

is called the **fundamental group** of X with base point x . For a map $f : X \rightarrow Y$ in $B(G)$, we put

$$G(X|Y) = \text{Aut}_Y(X).$$

f is called **Galois** if X and Y are connected and $G(X|Y)$ operates transitively on the fibres $f^{-1}(y)$.

Exercise 8. Let $f : X \rightarrow Y$ be a map of connected finite G -sets, and let $x \in X$, $y = f(x) \in Y$. Show that f is Galois if and only if $\pi_1(X, x)$ is a normal subgroup of $\pi_1(Y, y)$. In this case, one has a canonical isomorphism

$$G(X|Y) \cong \pi_1(Y, y)/\pi_1(X, x).$$

A pair of functors

$$A = (A^*, A_*) : B(G) \rightarrow (ab),$$

consisting of a contravariant functor A^* and a covariant functor A_* from $B(G)$ to the category (ab) of abelian groups is called a **double functor** if

$$A^*(X) = A_*(X) =: A(X)$$

for all $X \in B(G)$. We define

$$A_K = A(G/G_K).$$

If $f : X \rightarrow Y$ is a morphism in $B(G)$, then we put

$$A^*(f) = f^* \quad \text{and} \quad A_*(f) = f_*.$$

A homomorphism $h : A \rightarrow B$ of double functors is a family of homomorphisms $h(X) : A(X) \rightarrow B(X)$ representing natural transformations $A^* \rightarrow B^*$ and $A_* \rightarrow B_*$.

A **G -modulation** is defined to be a double functor A such that

(i) $A(X \sqcup Y) = A(X) \times A(Y)$.

(ii) If among the two diagrams

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \end{array} \quad \text{and} \quad \begin{array}{ccc} A(X) & \xrightarrow{g'^*} & A(X') \\ f_* \downarrow & & \downarrow f'_* \\ A(Y) & \xrightarrow{g^*} & A(Y') \end{array}$$

in $B(G)$, resp. (ab) , the one on the left is cartesian, then the one on the right is commutative.

Remark: G -modulations were introduced in a general context by A. DRESS under the name of **Mackey functors** (see [32]).

Exercise 9. G -modulations form an abelian category.

Exercise 10. If A is a G -module, then the function $A(G/G_K) = A^{G_K}$ extends to a G -modulation $\mathbf{A}$ in such a way that, for an extension $L|K$, the map $f^* : A_K \rightarrow A_L$, resp. $f_* : A_L \rightarrow A_K$, induced by $f : G/G_L \rightarrow G/G_K$, is the inclusion, resp. the norm $N_{L|K}$.

The rule $A \mapsto A$ is an equivalence between the category of G -modules and the category of G -modulations with “Galois descent”, i.e., of those G -modulations A such that

$$f^* : A(Y) \rightarrow A(X)^{G(X|Y)},$$

for every Galois mapping $f : X \rightarrow Y$, is an isomorphism.

Exercise 11. G -modulations are explicitly given by the following data. Let $B_0(G)$ be the category whose objects are the G -sets G/U , where U varies over the open subgroups of G , and whose morphisms are just the projections $\pi : G/U \rightarrow G/V$ for $U \subseteq V$, as well as the maps $c(\sigma) : G/U \rightarrow G/\sigma U \sigma^{-1}$, $\tau U \mapsto \tau U \sigma^{-1} = \tau \sigma^{-1}(\sigma U \sigma^{-1})$, for $\sigma \in G$.

Let $A = (A^*, A_*) : B_0(G) \rightarrow (ab)$ be a double functor and for $\pi : G/U \rightarrow G/V$ ($U \subseteq V$), resp. $c(\sigma) : G/U \rightarrow G/\sigma U \sigma^{-1}$ ($\sigma \in G$), define

$$\text{Ind}_V^U = A_*(\pi) : A(G/U) \rightarrow A(G/V),$$

$$\text{Res}_U^V = A^*(\pi) : A(G/V) \rightarrow A(G/U),$$

$$c(\sigma)_* = A_*(c(\sigma)) : A(G/U) \rightarrow A(G/\sigma U \sigma^{-1}).$$

If for any three open subgroups $U, V \subseteq W$ of G , one has the *induction formula*

$$\text{Res}_U^W \circ \text{Ind}_W^V = \sum_{U \cap W/V} \text{Ind}_U^{U \cap \sigma V \sigma^{-1}} \circ c(\sigma)_* \circ \text{Res}_{V \cap \sigma^{-1} U \sigma}^V,$$

then A extends uniquely to a G -modulation $A : B(G) \rightarrow (ab)$.

Hint: If X is an arbitrary finite G -set, then the disjoint union

$$A_X = \bigsqcup_{x \in X} A(G/G_x)$$

is again a G -set, because $c(\sigma)_* A(G/G_x) = A(G/G_{\sigma x})$. Define $A(X)$ to be the group

$$A(X) = \text{Hom}_X(X, A_X)$$

of all G -equivariant sections $X \rightarrow A_X$ of the projection $A_X \rightarrow X$.

Exercise 12. The function $\pi^{ab}(G/G_K) = G_K^{ab}$ extends to a G -modulation

$$\pi^{ab} : B(G) \rightarrow (\text{pro-}ab)$$

into the category of pro-abelian groups. Thus, for an extension $L|K$, the maps $f^* : G_K^{ab} \rightarrow G_L^{ab}$, resp. $f_* : G_L^{ab} \rightarrow G_K^{ab}$, induced by $f : G/G_L \rightarrow G/G_K$ are given by the transfer, resp. the inclusion $G_L \rightarrow G_K$.

Exercise 13. Let A be a G -modulation. For every connected finite G -set X , let

$$NA(X) = \bigcap f_* A(Y),$$

where the intersection is taken over all Galois maps $f : Y \rightarrow X$. Show that the function $NA(X)$ defines a G -submodulation NA of A , the **modulation of universal norms**.

Exercise 14. If A is a G -modulation, then the **completion** $\widehat{A}$ is again a G -modulation which, for connected X , is given by

$$\widehat{A}(X) = \varprojlim A(X)/f_* A(Y),$$

where the projective limit is taken over all Galois maps $f : Y \rightarrow X$.

For the following, let $d : G \rightarrow \widehat{\mathbb{Z}}$ be a fixed surjective homomorphism. Let $f : X \rightarrow Y$ be a map of connected finite G -sets and $x \in X$, $y = f(x) \in Y$. The **inertia degree**, resp. the **ramification index**, of f is defined by

$$f_{X|Y} = (d(G_y) : d(G_x)), \quad \text{resp.} \quad e_{X|Y} = (I_y : I_x),$$

where I_y , resp. I_x , is the kernel of $d : G_y \rightarrow \widehat{\mathbb{Z}}$, resp. $d : G_x \rightarrow \widehat{\mathbb{Z}}$. f is called **unramified** if $e_{X|Y} = 1$.

Exercise 15. d defines a G -modulation $\widehat{\mathbb{Z}}$ such that the maps f^* , f_* , corresponding to a mapping $f : X \rightarrow Y$ of connected G -sets, are given by

$$\widehat{\mathbb{Z}}(Y) = \widehat{\mathbb{Z}} \xrightleftharpoons[f_{X|Y}]{e_{X|Y}} \widehat{\mathbb{Z}} = \widehat{\mathbb{Z}}(X).$$

This gives a homomorphism of G -modulations

$$d : \pi^{ab} \longrightarrow \widehat{\mathbb{Z}}.$$

Exercise 16. An unramified map $f : X \rightarrow Y$ of connected finite G -sets is Galois, and d induces an isomorphism

$$G(X|Y) \cong \mathbb{Z}/f_{X|Y}\mathbb{Z}.$$

Let $\varphi_{X|Y} \in G(X|Y)$ be the element which is mapped to 1 mod $f_{X|Y}\mathbb{Z}$.

Let A be a G -modulation. We define a *henselian valuation* of A to be a homomorphism

$$v : A \rightarrow \widehat{\mathbb{Z}}$$

such that the submodulation $v(A)$ of $\widehat{\mathbb{Z}}$ comes from a subgroup $Z \subseteq \widehat{\mathbb{Z}}$ which contains $\mathbb{Z}$ and satisfies $Z/nZ = \mathbb{Z}/n\mathbb{Z}$ for all $n \in \mathbb{N}$. Let U denote the kernel of A .

Exercise 17. Compare this definition with the definition (4.6) of a henselian valuation of a G -module A .

Exercise 18. Assume that for every unramified map $f : X \rightarrow Y$ of connected finite G -sets, the sequence

$$0 \rightarrow U(Y) \xrightarrow{f^*} U(X) \xrightarrow{\varphi_{X|Y}^* - 1} U(X) \xrightarrow{f_*} U(Y) \rightarrow 0$$

is exact, and that $A(Y)^{[X:Y]} \subseteq f_*A(X)$ for every Galois mapping $f : X \rightarrow Y$ (the latter is a consequence of the condition which will be imposed in exercise 19). Then the pair (d, v) gives, for every Galois mapping $f : X \rightarrow Y$, a canonical “reciprocity homomorphism”

$$r_{X|Y} : G(X|Y) \rightarrow A(Y)/f_*A(X).$$

Exercise 19. Assume, beyond the condition required in exercise 18, that for every Galois mapping $f : X \rightarrow Y$ with cyclic Galois group $G(X|Y)$, one has

$$(A(Y) : f_*A(X)) = [X : Y] \quad \text{and} \quad \ker f_* = \text{im}(\sigma^* - 1),$$

where $[X : Y] = \#f^{-1}(y)$, with $y \in Y$, and σ is a generator of $G(X|Y)$. Then if $r_{X|Y}$ is an isomorphism for every Galois mapping $f : X \rightarrow Y$ of prime degree $[X : Y]$, so is

$$r_{X|Y} : G(X|Y)^{ab} \rightarrow A(Y)/f_*A(X),$$

for every Galois mapping $f : X \rightarrow Y$.

Exercise 20. Under the hypotheses of exercise 18 and 19 one obtains a canonical homomorphism of G -modulations

$$A \longrightarrow \pi^{ab}$$

whose kernel is the G -modulation NA of universal norms (see exercise 13). It induces an isomorphism

$$\widehat{A} \xrightarrow{\sim} \pi^{ab}$$

of the completion $\widehat{A}$ of A (see exercise 14).

Remark: The theory sketched above and contained in the exercises has a very interesting application to **higher dimensional class field theory**. In chap. V, (1.3), we will show that, for a Galois extension $L|K$ of local fields, there is a reciprocity isomorphism

$$G(L|K)^{ab} \cong K^*/N_{L|K}L^*.$$

The multiplicative group K^* may be interpreted in K -theory as the group $K_1(K)$ of the field K . The group $K_2(K)$ is defined to be the quotient group

$$K_2(K) = (K^* \otimes K^*)/R,$$

where R is generated by all elements of the form $x \otimes (1 - x)$. Treating Galois extensions $L|K$ of “2-local fields” — these are discretely valued complete fields with residue class field a local field (e.g., $\mathbb{Q}_p((x))$, $\mathbb{F}_p((x))((y))$) — the Japanese mathematician KAZUYA KATO (see [83]) has established a canonical isomorphism

$$G(L|K)^{ab} \cong K_2(K)/N_{L|K}K_2(L).$$

Kato’s proof is intricate and needs heavy machinery. It was simplified by the Russian mathematician I. FESENKO (see [36], [37], [38]). His proof may be viewed as a special case of the theory sketched above. The basic idea is the following. The correspondence $K \mapsto K_2(K)$ may be extended to a G -modulation K_2 . It does not satisfy the hypothesis of exercise 15, so that one may not apply the abstract theory directly to K_2 . But FESENKO considers on K_2 the finest topology for which the canonical map $(\ , \) : K^* \times K^* \rightarrow K_2(K)$ is sequentially continuous, and for which one has $x_n + y_n \rightarrow 0$, $-x_n \rightarrow 0$ whenever $x_n \rightarrow 0$, $y_n \rightarrow 0$. He puts

$$K_2^{\text{top}}(K) = K_2(K)/\Lambda_2(K),$$

where $\Lambda_2(K)$ is the intersection of all open neighbourhoods of 1 in $K_2(K)$, and he shows that

$$K_2^{\text{top}}(K)/N_{L|K}K_2^{\text{top}}(L) \cong K_2(K)/N_{L|K}K_2(L)$$

for every Galois extension $L|K$, and that $K_2^{\text{top}}(K)$ satisfies properties which imply the hypothesis of exercise 18 and 19 when viewing K_2^{top} as a G -modulation. This makes KATO’s theorem into a special case of the theory developed above.

§ 7. The Herbrand Quotient

The preceding section concluded abstract class field theory. In order to be able to apply it to the concrete situations encountered in number theory,

it is all important to verify the *class field axiom* (6.1) in these contexts. An excellent tool for this is the **Herbrand quotient**. It is a group-theoretic formalism, which we develop here for future use.

Let G be a finite cyclic group of order n , let σ be a generator, and A a G -module. As before, we form the two groups

$$H^0(G, A) = A^G / N_G A \quad \text{and} \quad H^{-1}(G, A) = N_G A / I_G A,$$

where

$$A^G = \{a \in A \mid a^\sigma = a\}, \quad N_G A = \left\{ N_G a = \prod_{i=0}^{n-1} a^{\sigma^i} \mid a \in A \right\},$$

$$N_G A = \{a \in A \mid N_G a = 1\}, \quad I_G A = \{a^{\sigma-1} \mid a \in A\}.$$

(7.1) Proposition. *If $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ is an exact sequence of G -modules, then we obtain an exact hexagon*

$$\begin{array}{ccccc} & & H^0(G, A) & \xrightarrow{f_1} & H^0(G, B) & & \\ & \nearrow f_6 & & & & \searrow f_2 & \\ H^{-1}(G, C) & & & & & & H^0(G, C) \\ & \nwarrow f_5 & & & \longleftarrow f_4 & & \\ & & H^{-1}(G, B) & & H^{-1}(G, A) & & \end{array}$$

Proof: The homomorphisms f_1, f_4 and f_2, f_5 are induced by $A \xrightarrow{i} B$ and $B \xrightarrow{j} C$. We identify A with its image in B so that i becomes the inclusion. Then f_3 is defined as follows. Let $c \in C^G$ and let $b \in B$ be an element such that $j(b) = c$. Then we have $j(b^{\sigma-1}) = c^{\sigma-1} = 1$ and $N_G(b^{\sigma-1}) = N_G(b^\sigma) / N_G(b) = 1$, so that $b^{\sigma-1} \in N_G A$. f_3 is thus defined by $c \bmod N_G C \mapsto b^{\sigma-1} \bmod I_G A$. In order to define f_6 , let $c \in N_G C$, and let $b \in B$ be an element such that $j(b) = c$. Then $j(N_G b) = N_G c = 1$, so that $N_G b \in A$. The map f_6 is now given by $c \bmod I_G A \mapsto N_G b \bmod N_G A$.

We now prove exactness at the place $H^0(G, A)$. Let $a \in A^G$ such that $f_1(a \bmod N_G A) = 1$; in other words, $a = N_G b$ for some $b \in B$. Writing $c = j(b)$, we find $f_6(c \bmod I_G C) = a \bmod N_G A$. Exactness at $H^{-1}(G, A)$ is deduced as follows: let $a \in N_G A$ and $f_4(a \bmod I_G A) = 1$, i.e., $a = b^{\sigma-1}$, with $b \in B$. Writing $c = j(b)$, we find $f_3(c \bmod N_G C) = a \bmod I_G A$. The exactness at all other places is seen even more easily. $\square$

(7.2) Definition. The **Herbrand quotient** of the G -module A is defined to be

$$h(G, A) = \frac{\#H^0(G, A)}{\#H^{-1}(G, A)},$$

provided that both orders are finite.

The salient property of the Herbrand quotient is its *multiplicativity*.

(7.3) Proposition. If $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ is an exact sequence of G -modules, then one has

$$h(G, B) = h(G, A) h(G, C)$$

in the sense that, whenever two of these quotients are defined, so is the third and the identity holds.

For a finite G -module A , one has $h(G, A) = 1$.

Proof: We consider the exact hexagon (7.1). Calling n_i the order of the image of f_i , we find

$$\begin{aligned} \#H^0(G, A) &= n_6 n_1, & \#H^0(G, B) &= n_1 n_2, & \#H^0(G, C) &= n_2 n_3, \\ \#H^{-1}(G, A) &= n_3 n_4, & \#H^{-1}(G, B) &= n_4 n_5, & \#H^{-1}(G, C) &= n_5 n_6, \end{aligned}$$

and thus

$$\begin{aligned} \#H^0(G, A) \cdot \#H^0(G, C) \cdot \#H^{-1}(G, B) \\ = \#H^0(G, B) \cdot \#H^{-1}(G, A) \cdot \#H^{-1}(G, C). \end{aligned}$$

At the same time, we see that if any two of the quotients are well-defined, then so is the third. And from the last equation, we obtain $h(G, B) = h(G, A)h(G, C)$. Finally, if A is a finite G -module, then the exact sequences

$$1 \longrightarrow A^G \longrightarrow A \xrightarrow{\sigma-1} I_G A \longrightarrow 1, \quad 1 \longrightarrow {}_{N_G} A \longrightarrow A \xrightarrow{N_G} N_G A \longrightarrow 1,$$

show that $\#A = \#A^G \cdot \#I_G A = \#{}_{N_G} A \cdot \#N_G A$, and $h(G, A) = 1$. $\square$

If G is an arbitrary group and g a subgroup, then to any g -module B , we may associate the so-called **induced G -module**

$$A = \text{Ind}_G^g(B).$$

It consists of all functions $f : G \rightarrow B$ such that $f(x\tau) = f(x)^\tau$ for all $\tau \in g$. The operation of $\sigma \in G$ is given by

$$f^\sigma(x) = f(\sigma x).$$

If $g = \{1\}$, we write $\text{Ind}_G(B)$ instead of $\text{Ind}_G^g(B)$. We have a canonical g -homomorphism

$$\pi : \text{Ind}_G^g(B) \longrightarrow B, \quad f \longmapsto f(1),$$

which maps the g -submodule

$$B' = \{f \in \text{Ind}_G^g(B) \mid f(x) = 1 \text{ for } x \notin g\}$$

isomorphically onto B . We identify B' with B . If g is of finite index, we find

$$\text{Ind}_G^g(B) = \prod_{\sigma \in G/g} B^\sigma,$$

where the notation $\sigma \in G/g$ signifies that σ varies over a system of left coset representatives of G/g .

Indeed, for any $f \in \text{Ind}_G^g(B)$ we have a unique factorization $f = \prod_{\sigma} f_{\sigma}^{\sigma}$, where f_{σ} denotes the function in B' which is determined by $f_{\sigma}(1) = f(\sigma^{-1})$.

If conversely A is a G -module with a g -submodule B such that A is the direct product

$$A = \prod_{\sigma \in G/g} B^\sigma,$$

then $A \cong \text{Ind}_G^g(B)$ via $B \cong B'$.

(7.4) Proposition. *Let G be a finite cyclic group, g a subgroup and B a g -module. Then we have canonically*

$$H^i(G, \text{Ind}_G^g(B)) \cong H^i(g, B) \quad \text{for } i = 0, -1.$$

Proof: Let $A = \text{Ind}_G^g(B)$ and let R be a system of right coset representatives for G/g with $1 \in R$. We consider the g -homomorphisms

$$\pi : A \longrightarrow B, \quad f \longmapsto f(1); \quad \nu : A \longrightarrow B, \quad f \longmapsto \prod_{\rho \in R} f(\rho).$$

Both admit the g -homomorphism

$$s : B \longrightarrow A, \quad b \longmapsto f_b(\sigma) = \begin{cases} b^\sigma & \text{for } \sigma \in g, \\ 1 & \text{for } \sigma \notin g, \end{cases}$$

as a section, i.e., $\pi \circ s = \nu \circ s = \text{id}$, and we have

$$\pi \circ N_G = N_g \circ \nu,$$

because one finds that, for $f \in A$,

$$(N_G f)(1) = \prod_{\tau \in g} \prod_{\rho \in R} f^{\rho\tau}(1) = \prod_{\tau} \prod_{\rho} f(\rho\tau) = \prod_{\tau} \left(\prod_{\rho} f(\rho) \right)^{\tau} = N_g(v(f)).$$

If $f \in A^G$, then $f(\sigma) = f(1)$ for all $\sigma \in G$, and $f(1) = f(\tau) = f(1)^{\tau}$ for all $\tau \in g$. The map π therefore induces an isomorphism

$$\pi : A^G \longrightarrow B^g.$$

It sends $N_G A$ onto $N_g B$, for one has $\pi(N_G A) = N_g(vA) \subseteq N_g B$ on the one hand, and on the other, $N_g(B) = N_g(v s B) = \pi(N_G(sB)) \subseteq \pi(N_G A)$. Therefore $H^0(G, A) = H^0(g, B)$.

As $N_g \circ v = \pi \circ N_G$, the g -homomorphism $v : A \rightarrow B$ induces a g -homomorphism

$$v : N_G A \longrightarrow N_g B.$$

It is surjective since $v \circ s = \text{id}$. We show that $I_G A$ is the preimage of $I_g B$. $I_G A$ consists of all elements $f^{\sigma^{-1}}$, $f \in A$, $\sigma \in G$. For if $G = (\sigma_0)$ and $\sigma = \sigma_0^i$, then $f^{\sigma^{-1}} = f^{(1+\sigma_0+\dots+\sigma_0^{i-1})(\sigma_0^{-1})} \in I_G A$. In the same way, one has $I_g B = \{b^{\tau^{-1}} \mid b \in B, \tau \in g\}$. Writing now $\sigma\rho = \rho'\tau\rho$, with $\rho, \rho' \in R$, $\tau\rho \in g$, we obtain

$$v(f^{\sigma^{-1}}) = \prod_{\rho \in R} \frac{f(\sigma\rho)}{f(\rho)} = \prod_{\rho'} \frac{f(\rho')^{\tau\rho}}{f(\rho')} = \prod_{\rho} b_{\rho}^{\tau\rho^{-1}} \in I_g B.$$

On the other hand, for $b^{\tau^{-1}} \in I_g B$, the function $f^{\tau^{-1}}$, with $f = sb$, is a preimage as $v(f^{\tau^{-1}}) = v s(b)^{\tau^{-1}} = b^{\tau^{-1}}$. After this it remains to show $\ker(v) \subseteq I_G A$. Let $G = (\varphi)$, $n = (G : g)$, $R = \{1, \varphi, \dots, \varphi^{n-1}\}$. Let $f \in N_G A$ be such that $v(f) = \prod_{i=0}^{n-1} \varphi^i = 1$. Define the function $h \in A$ by $h(1) = 1$, $h(\varphi^k) = \prod_{i=1}^{k-1} f(\varphi^i)$. Then $f(\varphi^k) = h(\varphi^k)/h(\varphi^{k-1}) = h(\varphi^{k-1})^{1-\varphi^{-1}}$ for $0 < k < n$, and $f(1)h^{\varphi^{-1}-1}(1) = \prod_{i=0}^{n-1} f(\varphi^i) = 1$. Hence $f = h^{1-\varphi^{-1}} \in I_G A$. Thus we finally get $H^{-1}(G, A) = H^{-1}(g, B)$. $\square$

Exercise 1. Let f, g be endomorphisms of an abelian group A such that $f \circ g = g \circ f = 0$. Make sense of the following statement. The quotient

$$q_{f,g}(A) = \frac{(\ker f : \text{im } g)}{(\ker g : \text{im } f)}$$

is multiplicative.

Exercise 2. Let f, g be two commuting endomorphisms of an abelian group A . Show that

$$q_{0,gf}(A) = q_{0,g}(A)q_{0,f}(A),$$

provided all quotients are defined.

Exercise 3. Let G be a cyclic group of prime order p , and let A be a G -module such that $q_{0,p}(A)$ is defined. Show that

$$h(G, A)^{p-1} = q_{0,p}(A^G)^p / q_{0,p}(A).$$

Hint: Use the exact sequence

$$0 \rightarrow A^G \rightarrow A \xrightarrow{\sigma-1} A^{\sigma-1} \rightarrow 0.$$

Let $N = 1 + \sigma + \cdots + \sigma^{p-1}$ in the group ring $\mathbb{Z}[G]$. Show that the ring $\mathbb{Z}[G]/\mathbb{Z}N$ is isomorphic to $\mathbb{Z}[\zeta]$, for ζ a primitive p -th root of unity, and that in this ring one has

$$p = (\sigma - 1)^{p-1} \varepsilon,$$

where ε is a unit in $\mathbb{Z}[G]/\mathbb{Z}N$.

Exercise 4. Let $L|K$ be a cyclic extension of prime degree. Using exercise 3, compute the Herbrand quotient of the group of units $\mathcal{O}_L^*$ of L , viewed as a $G(L|K)$ -module.

Exercise 5. If G is a group, g a normal subgroup and A a g -module, then $H^1(G, \text{Ind}_G^g(A)) \cong H^1(g, A)$.

